FORENSIC TECHNICAL AUDIT · AUDIT-ONLY PHASE

PAVEMENT-PINN

Comprehensive Forensic Technical Audit Report

Scientific Validity, PINN Mathematics, Software Architecture, Data Integrity, and Intel Edge AI Readiness

Physics-Informed Neural Networks for Intelligent Pavement Infrastructure Management
Intel AI Global Impact Festival 2026 — AI Changemakers

OVERALL AUDIT STATUS: FAIL — 5 OPEN P0 FINDINGS

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Document Control

Table 0.1 — Document metadata

Field	Value
Document ID	PAVEMENT-PINN-AUDIT-001
Title	PAVEMENT-PINN — Comprehensive Forensic Technical Audit Report: Scientific Validity, PINN Mathematics, Software Architecture, Data Integrity, and Intel Edge AI Readiness
Version	1.0 (initial issue)
Audit date	2026-10-14
Repository	`pavement-pinn` (local working copy)
Commit audited	`4a9b4fc49ec6e1d583648b2eaaa26293577add2e` — "feat: persist What-If runs; label legacy illustrative screens (G-SW-01, G-SW-04)"
Branch	`feature/m1-report-ownership` (local; not pushed per project notes)
Working-tree state	Uncommitted work of another session: 24 modified, 1 deleted, 10 untracked paths (demo-login and authentication rework: `server/auth/demoLogin.ts`, `src/features/authentication/*`, `tests/integration/auth-api.test.ts`, `tests/unit/demo-auth.test.ts`, docs). Not modified, staged or reverted by this audit. Code inspection and tests reflect this working tree.
Reference document	*Berikut adalah Master Blueprint & Kanvas Arsitektur Lengkap PAVEMENT-PINN (Versi Final Terbaru)*, 25 pages (BAB 1–10)
Other project sources consulted	`docs/SOURCE_INVENTORY.md` (spec v2.2, not available to this audit), `docs/validation/PINN_VALIDATION_GAP.md`, `docs/FINAL_AUDIT.md`, `docs/AI.md`, `docs/GAP_REGISTER.md`, `PROGRESS.md`
Classification	Internal engineering review — project team and mentors
Report status	Issued for review. Audit-only phase: no application code, formula, threshold, schema, data or configuration was changed.
Overall status	FAIL against the claim "validated PINN-based pavement engineering platform" (criteria below)
Prepared by	Automated audit session (Claude Code) at the request of the repository owner. No organisation, approval or endorsement is implied.

Table 0.2 — Revision history

Version	Date	Description
1.0	2026-10-14	Initial forensic audit: blueprint traceability, PINN identity, mathematical verification with an independent layered-elastic reference, data, tests, OpenVINO readiness, findings and roadmap

Audit scope. All tracked source under `src/`, `server/`, `tests/`, `scripts/`, `supabase/`, project documentation, and the uncommitted working tree as found; the master blueprint as the statement of intended requirements. Out of scope: sibling repositories in the parent folder, deployment hosts, external services, and the confidential contents of `docs/POST_STRUCTURE_AUDIT.md` (it locates credentials in other repositories and is deliberately not reproduced).

Audit limitations. (1) No Python runtime, model weights, datasets or OpenVINO installation exist on the audit machine, so training, CV evaluation and hardware benchmarks are BLOCKED. (2) The blueprint PDF could be read only through its text layer; some mathematical glyphs and the page design could not be inspected

visually, so the report's layout is reference-inspired, not a replica. (3) Spec v2.2, which the code follows, was not available to the audit. (4) The reference solver used for numerical verification was written during the audit and is verified against closed forms only. (5) The user's development server was not called; HTTP behaviour of legacy endpoints is inferred from code and function-level tests.

Evidence classification. VERIFIED directly demonstrated by inspection or a reproducible test; PARTIALLY VERIFIED partly demonstrated; NOT IMPLEMENTED absent from the inspected scope (the whole repository was searched unless stated); NOT VERIFIED insufficient evidence either way; FAILED a defined check was executed and failed; BLOCKED verification impossible because of a named dependency. Diagrams carry AS-IMPLEMENTED, BLUEPRINT REFERENCE or PROPOSED TARGET STATE. Implementation evidence, test evidence, scientific validation and deployment evidence are reported separately throughout.

Overall status criteria. PASS: no open P0 or P1; a trained physics-informed model with held-out validation; measured edge benchmarks. CONDITIONAL PASS: no open P0; every P1 has an accepted plan; every displayed AI, telemetry or engineering figure is either measured or adjacently labelled; the engine is either physically verified or explicitly positioned as a screening heuristic. FAIL: any open P0. **Rationale:** five P0 findings are open (F-01 to F-05): the engine is not a PINN, model and hardware telemetry are fabricated, strain responses contradict layered-elastic theory by one to two orders of magnitude, a fake training run is presented, and CV confidence is invented. Build and test success (226/226, clean type check) does not change this status.

Abbreviations and Terminology

Table 0.3 — Abbreviations used in this report

Term	Meaning
AC-WC / AC-BC / AC-Base	Asphalt Concrete Wearing / Binder / Base Course (Bina Marga)
BP-x	Blueprint requirement identifier (Appendix D)
CBR	California Bearing Ratio (%)
CV	Computer vision
ESAL / LEF	Equivalent Single Axle Load / Load Equivalency Factor
FEM	Finite Element Method
FWD	Falling Weight Deflectometer
IR	OpenVINO Intermediate Representation (.xml + .bin)
MLET	Multilayer linear-elastic theory (Burmister)
NNCF	Neural Network Compression Framework
NPU / iGPU	Neural Processing Unit / integrated GPU
ODOL	Over Dimension Over Loading
PCI	Pavement Condition Index (0–100)
PINN	Physics-Informed Neural Network
PMA / SMA / HRS	Polymer Modified Asphalt / Stone Matrix Asphalt / Hot Rolled Sheet
RBAC	Role-Based Access Control
RUL	Remaining Useful Life
SLM	Small Language Model
WIM	Weigh-In-Motion
εt / εv	Horizontal tensile strain at the bottom of the asphalt / vertical compressive strain at the top of the subgrade (με)
F-xx / S-xx / R-xx / T-xx	Finding / strength / remediation task / test identifiers

Executive Summary

1. Implementation maturity. PAVEMENT-PINN is a working web prototype (React 19 + Vite, Express API, JSON file store) with a well-engineered, test-covered multi-stakeholder workflow: anonymous citizen reporting, operations review and assignment, field inspection, recommendation, approval with separation of duties, internal work orders, verification and an audit trail (226 automated tests pass; type check clean). The scientific and AI components that give the project its name are not implemented: there is no neural network, no training, no CV model, no OpenVINO runtime and no measured benchmark.

2. Scientific validity of the physics engine. The engine (`src/engines/pinn/core/pinnEngine.ts`) is a set of independent power-law formulas with literal coefficients. An independent multilayer linear-elastic reference solver written for this audit (verified against the Boussinesq closed form to ≈ 1e-12 and against Burmister two-layer chart values) shows the engine's tensile strains at a median of about 5–7 % and compressive strains at about 2–2.5 % of elastic theory across the documented input range. For the blueprint's default segment (AC-WC 10 cm, 2500 MPa, CBR 6) under a 10 t axle the engine gives εt = 20 με and εv = 17 με where the reference gives about 300–435 με and 790–1170 με. The engine therefore reports typical structures as "STABIL … dalam batas elastis aman" when elastic theory places them well above common fatigue and rutting limits.

3. Engine identity. Deterministic analytical heuristic. It is not a trained PINN, not a surrogate of any reference model, and not a hybrid. The newer domain layer already labels it `RULE_BASED`; several legacy screens, the README title and the exported PDF still present it as a PINN running on OpenVINO.

4. Physics equations. Equilibrium, constitutive law, boundary conditions and interface continuity (BAB 5.1) are not implemented or enforced anywhere in executable code. The engine fails the elastic-similarity test that every layered-elastic solution satisfies (exponents −0.4/−0.75/−0.6 instead of −1).

5. Reference data. None. No FWD, FEM, WIM, core, CV image or longitudinal PCI data is in the repository; segment attributes are hand-authored synthetic values, and some screens contradict that with "FWD Validated" and WIM-station badges.

6. Integration status. Workflow, RBAC (demo identity), persistence (local atomic file store) and What-If persistence: implemented and tested. CV: keyword matching with constant confidence (DEMO; human review always required on the server). Weather and traffic: deterministic demo providers, honestly labelled on new surfaces. Supabase: migration written, not applied. Pilot corridor: Karawang (spec v2.2), not the blueprint's Jalan Raya Narogong.

7. Edge inference. No verified capability. All latency, power, NPU utilisation, quantisation-accuracy and loss figures in the UI are constants or random numbers; the "OpenVINO IR export" is a hand-written XML without weights; the hardware detector claims an Intel NPU by default.

8. Critical blockers. (a) The identity and telemetry misrepresentations (F-01, F-02, F-04, F-05); (b) the physically inconsistent response model (F-03) and its downstream life and threshold logic (F-06, F-07); (c) the absence of any empirical dataset (F-16); (d) no Python/ML toolchain (G-PY-01); (e) an unresolved conflict between the blueprint and the spec the code follows (F-15).

9. Remediation priorities. First remove or relabel every unsupported claim and every random or constant "measurement" (Phase P0, Chapter 13); then replace the response formulas with a verified layered-elastic computation and unify thresholds and the standard axle; then connect persistence and real data providers; only then train and validate a PINN or surrogate and benchmark it on Intel hardware.

Table 0.4 — Executive summary by domain

Domain	Blueprint requirement	Actual evidence	Status	Risk	Next action
PINN identity	Trained PINN with data, physics, BC and IC losses (BAB 5.1)	One closed-form function; no network, weights, optimiser or residual	NOT IMPLEMENTED	Critical	Relabel (R-P0-01); decide model form (R-P0-04)
Physics consistency	Burmister multilayer elasticity	Strains 1–2 orders below MLET reference; similarity test fails	FAILED	Critical	Verified MLET computation (R-P1-01)
Thresholds and failure	εt, εv limits leading to life prediction	180/200/250/350 με uncited and inconsistent; no transfer functions	FAILED	High	Threshold register + one criterion (R-P1-02)
RUL / PCI / recommendation	Days to failure, PCI trend, PMA vs AC-WC	Heuristics independent of structure; overlay = PCI look-up	FAILED	High	Deterioration model (R-P3-04)
Data	FWD/FEM data loss; GRDDC + local CV data; SNI baseline	No empirical data; synthetic attributes	NOT IMPLEMENTED	Critical	Data acquisition (R-P2-04)
Computer vision	YOLOv8 + anti-spoofing	Keyword match, constant confidence; no anti-spoofing	NOT IMPLEMENTED	Critical	Remove claims (R-P0-02); model (R-P3-05)
Weather / traffic	BMKG weather; virtual WIM from traffic API	Deterministic demo providers; illustrative axle spectrum	PARTIALLY VERIFIED	Medium	Live adapters (R-P2-03)
Workflow and RBAC	Four roles, approvals, repair verification	Server-enforced state machine, audit trail, tests pass	VERIFIED (demo identity)	Low	Real identity provider (R-P2-02)
Persistence	Supabase PostgreSQL + Storage + Realtime	Local JSON store; migration not applied	BLOCKED	Medium	Apply migration (R-P2-01)
OpenVINO / edge	ONNX → IR → NNCF; < 10 ms on CPU/iGPU/NPU	Simulated telemetry; illustrative IR; no runtime	NOT IMPLEMENTED	Critical (claim)	Remove figures (R-P0-02); benchmark after model (R-P3-06)
Geospatial twin / UI	3D isometric map, Three.js cross-section, realtime pins	2D maps, 2D cross-section, no realtime	PARTIALLY VERIFIED	Medium	Decide scope (R-P0-05)
GenAI	On-device Phi-3/Llama-3.2 on NPU	Cloud Gemini call + template fallback	DIVERGENT	Medium	Decide and document (R-P0-05)

Chapter 1 — Project Context and Blueprint Traceability

1.1 Problem definition, pilot corridor and SDGs

The blueprint (BAB 1) motivates the system with national road-condition and budget statistics, the ODOL problem and the fourth-power law, and selects **Jalan Raya Narogong (Bekasi–Bogor)** as the Phase 1 pilot, with Karawang (Klari–Simpang Jomin) and Semarang as Phase 2 expansion. The implementation follows a later specification, *PAVEMENT-PINN v2.2* (`docs/SOURCE_INVENTORY.md`), whose pilot is **Karawang Pantura Klari → Jatisari → Simpang Jomin** (`src/data/reference/karawangPilot.ts`, KRW-001..004). The string "Narogong" does not occur in `src/`; a report near Narogong (approximately −6.38, 107.00, an audit approximation) is flagged `OUT_OF_SCOPE: di luar koridor pilot Klari–Jatisari` by `screenReport` (`src/domain/workflow/screening.ts:40,83`) VERIFIED. The statistics in BAB 1.1 were not re-verified (no source documents supplied); several of them reappear in policy screens as constants (F-22). SDG alignment (BAB 1.3) is narrative N/A.

> **Source-of-truth conflict (F-15).** The audit reference ("Versi Final Terbaru") and the specification implemented in code disagree on the pilot corridor, the technology stack, the 3D twin and the on-device SLM. Until the team records which document governs, every traceability row marked DIVERGENT is a decision, not a defect.

1.2 Research gap and ten dimensions of novelty

Of the ten novelty dimensions (BAB 2.2), one is implemented and verified at software level (N6, one computation feeding three role views), one is verified for a demo identity model (N7, RBAC), two are partially present (N1, N3), and six are not implemented (N2 PINN losses, N4 virtual WIM and anti-spoofing, N5 recommender and quality audit, N8 heterogeneous Intel execution, N9 NNCF quantisation, N10 on-device SLM — the last implemented differently, via a cloud model). Rows BP-2.2-N1 to N10 in Appendix D give the evidence.

1.3 End-to-end pipeline (BAB 3) and system flow (BAB 9)

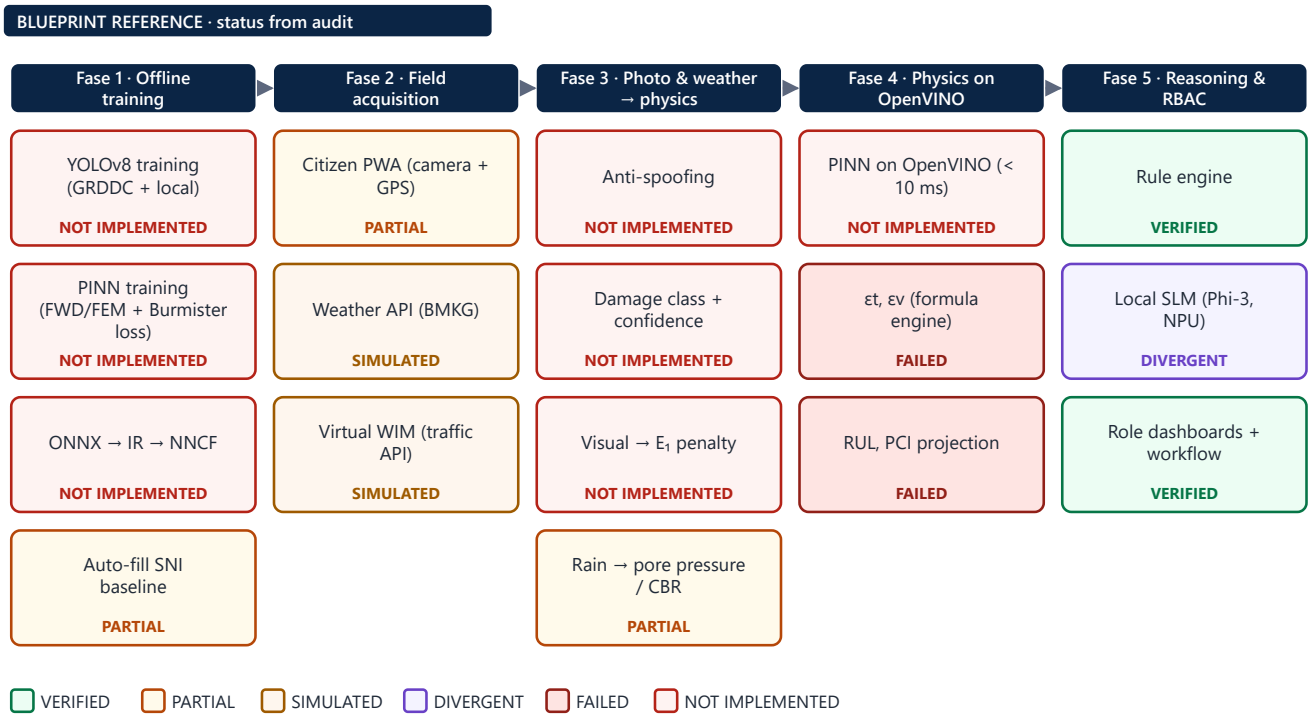


Figure 1.1 — Blueprint five-phase pipeline (BAB 3) with the implementation status of each stage. BLUEPRINT REFERENCE, annotated with audit status.

Phase 1 (offline training and compression) is entirely absent. Phase 2 (field acquisition) is present for the citizen report only; weather and traffic are demo providers. Phase 3 (translation from photo and weather to physics) is deliberately not implemented in the new domain layer — correctly, because no calibration exists — while the legacy engine applies uncited linear moisture and temperature factors. Phase 4 runs the formula engine instead of a PINN on OpenVINO. Phase 5 has a rule engine and a deterministic explanation layer but no local SLM.

1.4 Chapters 4–8 and 10 of the blueprint

Material catalogue, ageing, recommender and sub-standard audit (BAB 4): catalogue present as static reference text only; the others are not implemented (F-18, F-26). PINN and OpenVINO (BAB 5): not implemented (Chapters 3 and 10). Stakeholder roles (BAB 6): implemented with one deliberate divergence — the Field Engineer recommends and the Operations Manager approves, and the "SPK" is an internal work order explicitly marked as not an official document; the audit regards this separation of duties as an improvement that should be written back into the blueprint. 3D twin (BAB 7): 2D maps (Leaflet, MapLibre without pitch), a 2D stacked-`div` cross-section, no Three.js. Architecture and schema (BAB 8): different stack; the Supabase schema normalises the blueprint tables into `report_cases`, `damage_detections`, `pinn_predictions` and `simulations`, which is defensible but unapplied. Evaluation (BAB 10.1): none of the accuracy, latency or usability metrics has been measured.

1.5 Requirements traceability summary

The full matrix (67 requirements) is in Appendix D. Counted totals: 3 VERIFIED, 24 PARTIALLY VERIFIED, 7 DIVERGENT, 26 NOT IMPLEMENTED, 4 NOT VERIFIED, 3 N/A. Every VERIFIED row concerns workflow software; no scientific or AI requirement is verified.

Table 1.1 — Representative traceability rows (full matrix in Appendix D)

Requirement ID	Blueprint reference	Expected behaviour	Implementation location	Evidence	Status	Gap
BP-1.2a	BAB 1.2	Pilot Jalan Raya Narogong	`karawangPilot.ts`; `screening.ts:40`	Narogong report OUT_OF_SCOPE	DIVERGENT	F-15
BP-3.F1b	BAB 3 Fase 1	PINN trained on FWD/FEM + Burmister loss	—	No training code	NOT IMPLEMENTED	F-01
BP-3.F3b	BAB 3 Fase 3	YOLOv8 class + confidence	`cvDamageClasses.ts:216-244`	Keyword match	NOT IMPLEMENTED	F-05
BP-3.F4a	BAB 3 Fase 4	εt, εv from PINN on OpenVINO	`pinnEngine.ts:58-69`	Formula, 1-2 orders low	PARTIALLY VERIFIED	F-03
BP-6.0	BAB 6	Four roles with RBAC	`domain/workflow`, `features/rbac`	Tests pass	VERIFIED	IdP BLOCKED
BP-7.2d	BAB 7.2	Three.js 3D cross-section	`SimulationPanel.tsx:460-517`	2D divs	NOT IMPLEMENTED	F-15
BP-10.1a	BAB 10.1	RMSE/MAE/R^2 > 0.90 vs FWD/FEM	—	No evaluation	NOT IMPLEMENTED	F-16

Chapter 2 — Repository and Software Architecture Audit

2.1 Repository inventory

222 tracked files (excluding lock files: about 58 000 lines, of which the golden fixture is 9 885). Main areas: `src/app` (shell, providers, layouts; `ModernAppLayout.tsx` alone 2 820 lines), `src/features` (role views, maps, policy, WIM, work orders), `src/domain` (workflow state machine, intelligence: assessment, ESAL, evidence providers, What-If, Road Story), `src/engines` (the formula engine and the CV class table), `src/runtime` (simulated hardware detection and benchmark), `server` (Express app, routes, controllers, services, file store), `supabase/migrations` (one migration, not applied), `tests` (24 files, 226 tests). There is no Python code, no notebook, no model file and no dataset other than TypeScript constants.

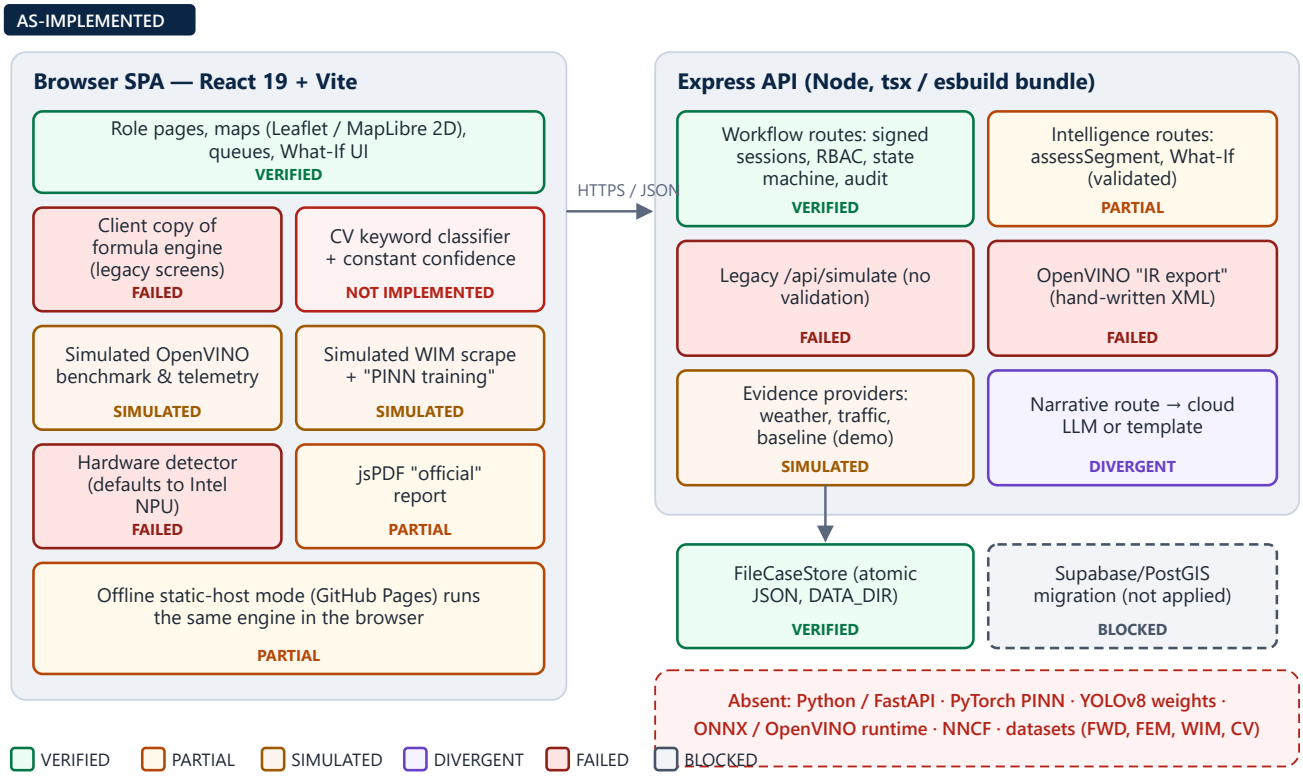


Figure 2.1 — As-implemented architecture and data flow. AS-IMPLEMENTED (from source inspection).

2.2 Critical execution paths

Table 2.1 — Classification of critical features

Feature	Path (input → output)	Classification	Evidence
Citizen report submission	UI form → `POST /api/workflow/reports` → `checkPhotoReference` → `screenReport` (DEMO CV) → `FileCaseStore.commit`	1. Fully implemented (demo identity)	workflow tests; `photoValidation.ts`; `fileCaseStore.ts:44-45`

Feature	Path (input → output)	Classification	Evidence
Review → assign → inspect → recommend → approve → work order → verify → close	`applyCommand` with role and version guards	1. Fully implemented	"cross-role golden path" test
Engineering assessment	`GET /api/intelligence/...` → `assessSegment` → `runPinnSimulation` (clamped)	4. Deterministic calculation presented, on new surfaces, honestly as RULE_BASED	`assessment.ts:114-196`
Legacy simulation	`POST /api/simulate` → `runPinnSimulation` (no validation)	4. Deterministic calculation presented as PINN + OpenVINO	`simulationController.ts`; F-01, F-11
What-If	`POST` simulation route → `runWhatIf` → persisted SIMULATED record	2. Implemented with limitations (engine F-03)	`intelligence.routes.ts:95-125`
CV damage classification (UI)	photo URL/data URL → `analyzeImageDamageClass` → constant confidence	3. Mocked	`cvDamageClasses.ts:216-244`
Weather, traffic → load	`DEMO_WEATHER_PROVIDER`, `estimateTrafficLoad`	3. Mocked (labelled DEMO / INFERRED)	`evidence.ts`
OpenVINO telemetry and benchmark	constants + `Math.random` / JS loop	3. Mocked, displayed as measurement	F-02
WIM scraping + PINN training modal	`Math.random` records → fixed loss multiples	3. Mocked, displayed as real	F-04
GenAI narrative	`POST` narrative route → cloud model or template	2. Implemented with limitations; not verified at runtime	`narrativeService.ts`
PDF report	`generateOfficialReportPdf` (jsPDF, client)	2. Implemented; content inherits F-02/F-03	F-19
Supabase persistence	migration only	6. Not verified (not applied)	F-24

2.3 Code-quality observations

Dead or disconnected code: `src/data/mock/auditLogs.ts` (`INITIAL_AUDIT_LOGS`, not imported; contains "Regangan Tarik 245 με"). Duplicate implementations: two strain-limit charts (`FieldEngineerTools.tsx`, `FieldEngineerDashboard.tsx`) with different legends for the same field; two fourth-power bases (F-08). Fallback behaviour that produces plausible numbers when no result exists: strain placeholders 245/380 με, loss placeholders 0.00342/0.00185/0.00157, latency "6.8 ms". Swallowed errors: the hardware detector's `catch { webGlSupported = false }` keeps the Intel defaults (F-12). Misleading UI state: a hard-coded "MELEBIHI AMBANG BATAS (+22.5%)" banner (F-06). Discrepancy between tests and runtime: tests pin the formulas' current output; nothing tests their physics. Strengths are listed as S-01 to S-08 in Appendix A.3.

2.4 Authentication, authorisation and security (summary)

Server-signed session tokens; actor derived from the token, never the body; role, assignment, ownership and separation-of-duty rules enforced in the domain service; official-only audit access; public projection hides staff identities; magic-byte upload validation; rate limiting in process. Gaps: no identity provider (demo accounts, **BLOCKED** on credentials), no EXIF/GPS stripping, the six original endpoints (`/api/simulate`, `/api/roads`, `/api/openvino/*`, narrative, health) are unauthenticated, and `/api/simulate` is unvalidated (F-11). The uncommitted demo-login rework was observed but not audited in depth because it is in progress in another session.

Chapter 3 — Forensic PINN Identity and Training Audit

3.1 Method

The audit searched the whole repository for every element a PINN requires: model definitions (classes, layer arrays, weight tensors), automatic differentiation, residual computation, loss terms, optimisers, training loops, datasets, checkpoints and model loaders, plus dependencies (`package.json`) and non-TypeScript sources. It then traced every call site of the engine (`runPinnSimulation` is called from eight production modules: `AppStateProvider`, `ModernAppLayout`, `SimulationPanel`, `ExecutivePolicyView`, `StrategicPolicyMapView`, `MyTasksView`, `server/services/simulationService` and `domain/intelligence/assessment`).

3.2 Model identity

Table 3.1 — Model identity register

Property	Required evidence	Observed implementation	Finding
Neural architecture	Model definition (layers, activations)	None. `src/engines/pinn/core/pinnEngine.ts` is one exported function of 296 lines	NOT IMPLEMENTED
Trainable parameters	Parameter inventory	None; every coefficient is a literal (120, 1.75, 0.4, 210, 1.4, 0.75, 45, 0.6, 0.5, 0.0018, …)	NOT IMPLEMENTED
Forward pass	Network evaluation	Closed-form expressions EQ-4 to EQ-11	Deterministic formula
Automatic differentiation	Gradient of outputs w.r.t. coordinates	None; no spatial coordinates are inputs	NOT IMPLEMENTED
Physics residual	PDE residual at collocation points	None. `physicsLossValue = 0.0018 + Math.random()*0.0006`	NOT IMPLEMENTED; displayed value fabricated (F-02)
Data loss	Comparison with FWD/FEM data	None. `dataLossValue = 0.0011 + Math.random()*0.0004`	NOT IMPLEMENTED; fabricated (F-02)
Boundary / initial-condition loss	BC and IC terms	None (the WIM modal derives a "boundaryLoss" as `dataLoss × 0.42 + 0.00085`)	NOT IMPLEMENTED; fabricated (F-04)
Loss weighting, optimiser	λ weights; Adam + L-BFGS	None; the text "Konvergensi Adam Optimizer epoch 1500." is a constant	NOT IMPLEMENTED
Training loop	Gradient updates over data	None in code; the "training" modal runs three `setTimeout` steps	NOT IMPLEMENTED
Training data and splits	Dataset with provenance, train/val/test	None	NOT IMPLEMENTED
Checkpoint	Reproducible trained weights	None in repository or `dist/`	NOT IMPLEMENTED
Serialisation / loading	Model file and loader	None; the IR "export" is hand-written text	NOT IMPLEMENTED (F-13)
Inference	Loaded model execution	Function call with clamped inputs (new layer) or raw inputs (legacy)	Deterministic
Reproducibility	Same input → same output	Strains yes; telemetry and ids no	PARTIALLY VERIFIED (F-21)

Property	Required evidence	Observed implementation	Finding
Provenance	Model version recorded with outputs	New layer: `pinnEngine-formula@1 (deterministic, no trained weights)`	VERIFIED (new layer only)

BLUEPRINT REFERENCE (BAB 5.1) vs AS-IMPLEMENTED

Element	Required for a PINN	Present in repository
Neural network u_θ (layers, activations)	yes	absent
Trainable parameters θ	yes	absent
Spatial coordinates as inputs	yes	absent
Automatic differentiation of u_θ	yes	absent
PDE residual at collocation points	yes	absent
Boundary / interface condition loss	yes	absent
Data loss vs FWD / FEM	yes	absent
Optimiser (Adam + L-BFGS), training loop	yes	absent
Training data with split	yes	absent
Checkpoint and model loader	yes	absent
Closed-form formulas with literal coefficients	no	present
Threshold rules (alerts, notes, overlay)	no	present

Classification: deterministic analytical heuristic — not a PINN, not a surrogate, not a hybrid.

Figure 3.1 — Required PINN elements versus elements present. Left: BLUEPRINT REFERENCE (BAB 5.1). Right: AS-IMPLEMENTED.

3.3 Conclusion on identity

The engine is a **deterministic analytical heuristic**. It is not a PINN (no network, no physics residual), not a surrogate (it was not fitted to any reference solution present in the repository, and it disagrees with layered-elastic theory by one to two orders of magnitude), and not a hybrid. Training has not occurred: no artifact, log, dataset or code demonstrates it, and the existence of words such as "PINN", "training" or "loss" in UI text is not evidence of training. A deterministic engine can be a legitimate screening tool, but only if it is physically consistent and described as what it is; neither condition holds today for the legacy screens (F-01, F-03).

Chapter 4 — Mathematical Audit of Pavement Physics

4.1 Multilayer elastic theory

The blueprint specifies Burmister multilayer elasticity with equilibrium and an isotropic constitutive law:

$$\nabla \cdot \sigma + \mathbf{b} = 0 \quad \text{(EQ-2)}$$

Equilibrium (BLUEPRINT REFERENCE, BAB 5.1).

$$\sigma = \mathrm{C} : \varepsilon \quad \text{(EQ-3)}$$

Constitutive law (BLUEPRINT REFERENCE, BAB 5.1).

Neither relation is evaluated in code, and the engine's inputs cannot support them: there is no field variable, no spatial coordinate, no Poisson ratio, no contact pressure or radius, and only one structural layer plus a subgrade (Table 4.1). An equilibrium residual alone would also not close the problem: Burmister's solution additionally needs the constitutive law for each layer, strain–displacement compatibility, the surface traction condition under the tyre, interface continuity (bonded or frictionless) and decay at depth and distance — the boundary-value problem solved by the audit's reference (EQ-12, Appendix B.4).

Table 4.1 — Mechanical-model elements required by MLET vs present in the engine

Element	Required by MLET	Present in engine	Status
Equilibrium	yes	no	NOT IMPLEMENTED
Stress–strain law per layer (E, ν)	yes	E only, no ν	NOT IMPLEMENTED
Strain–displacement compatibility	yes	no	NOT IMPLEMENTED
Layer geometry	all layers incl. base, sub-base	asphalt thickness; a hidden "+15 cm" in εv	PARTIALLY VERIFIED
Surface load	pressure q over radius a, wheel configuration	axle tonnes only; "surface stress" $0.65P$ / 10 MPa	FAILED (F-09)
Interface conditions	continuity of $u_z, u_r, \sigma_z, \tau_{rz}$	none; responses read one layer each	FAILED
Conditions at depth / distance	decay	basin $w_0 e^{-0.0018r}$ fixed for every structure	FAILED
Assumptions (linear, isotropic, homogeneous layers, quasi-static)	stated	implicit, not stated	NOT VERIFIED

The engine's responses are:

$$\varepsilon_t = \mathrm{round}\left(\frac{120P}{h^{1.75} \left(E_1' / 2000 \right)^{0.4}} \right) \quad \text{(EQ-4)}$$

Engine tensile strain (AS-IMPLEMENTED, pinnEngine.ts:58-62).

$$\varepsilon_v = \mathrm{round}\left(\frac{210P}{\left(h + 15 \right)^{1.4} \left(E_s / 40 \right)^{0.75}} \right) \quad \text{(EQ-5)}$$

Engine compressive strain (AS-IMPLEMENTED, pinnEngine.ts:65-69).

Because every linear-elastic layered solution satisfies the similarity property (EQ-14), multiplying all moduli by k must divide all strains and displacements by k. Applying the same scaling to EQ-4 to EQ-6 gives exponents −0.4, −0.75 and −0.6 (EQ-15). The engine therefore cannot be a layered-elastic solution or an approximation of one with uniform accuracy FAILED.

4.2 Numerical comparison with an independent reference

The audit implemented an axisymmetric MLET solver (Love strain function, Hankel transform, bonded interfaces; `evidence/mlet.mjs`) and verified it before use (Table B.2): a homogeneous three-layer system reproduces the Boussinesq closed form to ≈ 1e-12 relative error at all tested depths; strains from displacement and stress kernels agree to 1e-14 in a strongly layered system; and Burmister two-layer interface stresses match published chart values (0.646, 0.291, 0.081 for modular ratios 1, 10, 100). It was then compared with the engine over 144 in-range cases.

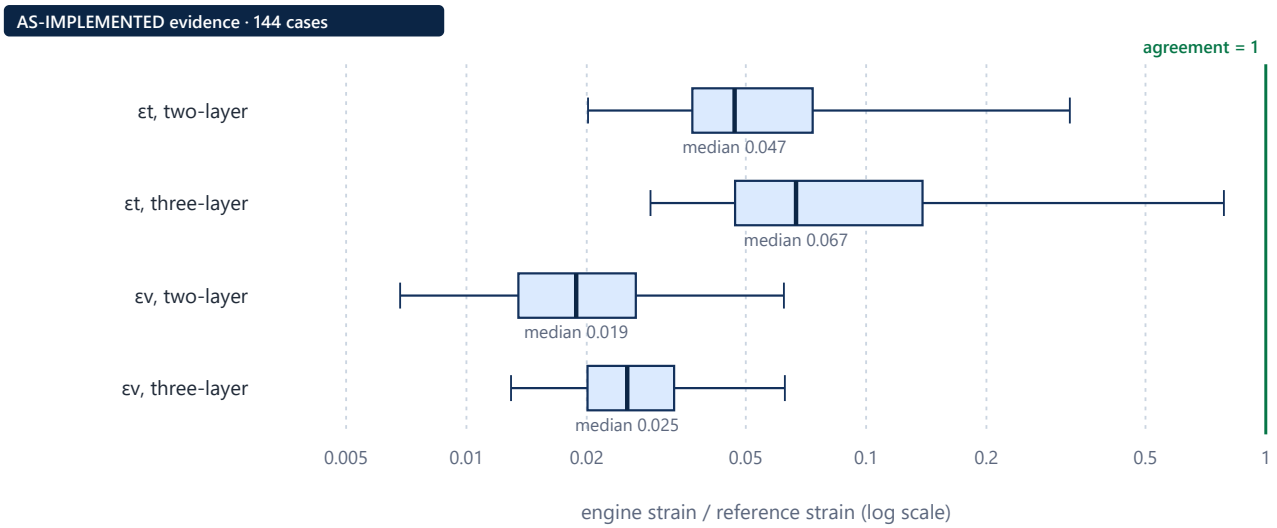


Figure 4.1 — Distribution of engine/reference strain ratios over the 144-case grid (box: inter-quartile range; whiskers: min–max). A ratio of 1 means agreement. AS-IMPLEMENTED evidence.

Results (Tables B.3–B.5): median ratios 0.047 (εt, two-layer), 0.067 (εt, three-layer), 0.019 and 0.025 (εv). Sensitivities differ qualitatively: in the engine εt does not depend on subgrade stiffness and εv does not depend on asphalt stiffness; in the reference both do. Load enters linearly in the engine but with exponent ≈ 0.6 in the reference at constant tyre pressure. The finding is robust to the loading assumptions (B.4) and classified VERIFIED against the reference; empirical accuracy of both remains NOT VERIFIED for lack of field data.

4.3 Dimensional analysis

EQ-4 to EQ-6 combine tonnes, centimetres and megapascals through fractional exponents, so their coefficients carry hidden units (for EQ-4, με·cm^1.75·t^{-1}) and the expressions are not dimensionally homogeneous; their correctness could only come from calibration, and no calibration record exists. Specific issues: axle load is never converted to force or to a wheel load; "surface stress" scales with axle load instead of being set by tyre pressure; the εv formula adds an unexplained 15 cm; $E_s = 10$ CBR MPa is an uncited correlation applied for CBR 3–10 %; strain is rounded to integer microstrain; rainfall intensity (mm/h) is used directly as a moisture state; the three time quantities (PCI horizon "per year", RUL years, pothole horizon days) are not linked by a common time model; PCI is used as a projection variable although it is defined by distress surveys. Full unit trace: Table B.8.

4.4 Boundary and edge cases

Function-level tests (Table B.7) show non-finite results for zero load (RUL and horizon ∞, Zero-ODOL impact −∞), zero thickness, zero modulus and zero or negative CBR; negative strains for negative loads; silent `NaN` propagation with a GREEN alert; string coercion; negative Zero-ODOL percentages below 10 t; unbounded linear stiffening below 25 °C; and saturation that masks absurd inputs (1000 mm/h). The new `assessSegment` path clamps inputs to the documented ranges and flags them (F-28); the legacy `/api/simulate` path performs no validation (F-11). No equation was modified.

Chapter 5 — Strain Thresholds and Failure-Criterion Audit

Every strain-related threshold was traced (Table B.6). Four numeric limits exist for two quantities — 180, 200 and 250 με for εt, 350 με for εv — plus fabricated placeholder values (245, 380 με) and a hard-coded "+22.5 %" exceedance banner. None cites a document, clause, table or calibration; the same field is labelled "AASHTO" in one chart and "Bina Marga" in another; and the engine's KRITIS rule triggers at 250 με while its message cites a 200 με limit FAILED.

Table 5.1 — Strain thresholds (abridged from Table B.6)

Parameter	Source location	Current value	Physical meaning	Source/standard	Verification status	Risk
εt warning	`pinnEngine.ts:130`	180 με	approaching fatigue	none	UNVERIFIED	Medium
εt limit	`pinnEngine.ts:129`, `FieldEngineerTools.tsx:198`, `pdfGenerator.ts:111`	200 με	fatigue limit	"AASHTO" / "Bina Marga" (uncited, inconsistent)	UNVERIFIED	High
εt critical	`pinnEngine.ts:128`	250 με	exceeds limit	none; contradicts own text	FAILED	High
εv limit	`FieldEngineerTools.tsx:203`	350 με	rutting limit	none	UNVERIFIED	High

Engineering distinctions the implementation does not make:

- **Tensile strain at the bottom of the asphalt** governs bottom-up fatigue cracking; **vertical compressive strain at the top of the subgrade** governs structural rutting. The code computes both but evaluates neither against a number of load repetitions.
- **Elastic response versus accumulated permanent deformation**: all outputs are single-load elastic responses; no accumulation model exists.
- **Fatigue life versus rutting life**: no life is computed from strain at all — RUL and pothole horizon come from load factor and rain only (EQ-10).
- **Structural failure versus visible pothole formation**: the "pothole horizon" is a rule, not a failure criterion, and potholes also depend on surface ravelling, water ingress and patch quality, which no input describes.

Reachability: on the 144-case grid the engine exceeds 200 με in 3 cases and never exceeds 350 με (maximum 98 με), whereas the reference exceeds them in 111–123 and 126–134 cases respectively. The thresholds and the response scale are therefore mutually inconsistent; with a physically consistent response model, the fixed limits would be exceeded in most cases, which shows that a fixed limit without design traffic is not meaningful either. Required to close: the governing standard and clause for each limit chosen by the team (the audit does not substitute one), or transfer functions of the form EQ-16 with coefficients from that standard, evaluated with design ESAL (R-P1-02).

Chapter 6 — Physics-to-Application Translation Audit

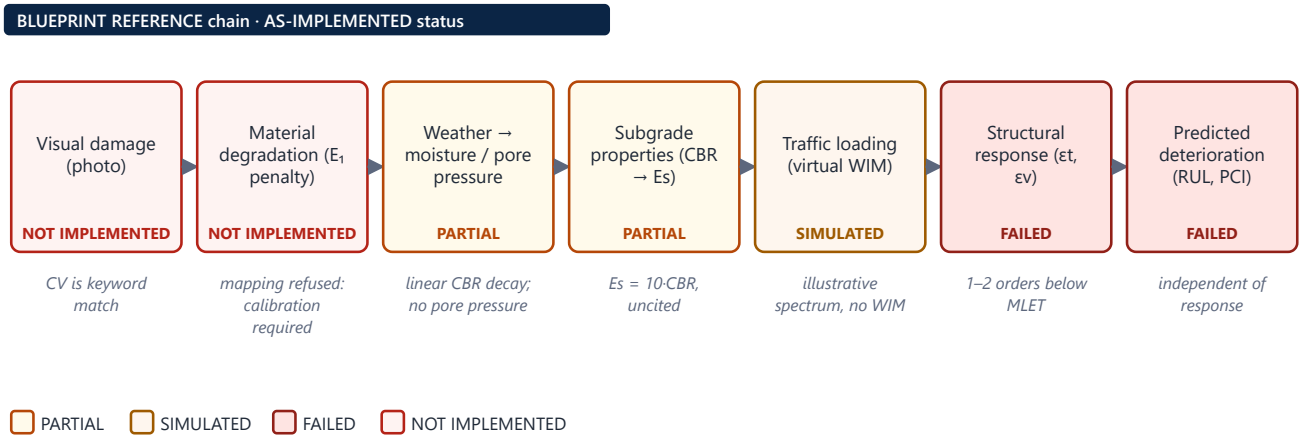


Figure 6.1 — Translation chain from observation to predicted deterioration, with the implementation status of each link. BLUEPRINT REFERENCE chain, AS-IMPLEMENTED status.

Table 6.1 — Status of each translation step

Step (blueprint)	Implementation	Nature of output	Assessment
Visual damage → material degradation (E_1 penalty 40–60 % for severe alligator cracking)	New layer refuses the mapping (`translateSurfaceEvidence`, status CALIBRATION_REQUIRED); legacy engine ignores damage class	Surface indicator only	Correct refusal. A CV label is not a modulus measurement; a back-calculated modulus needs FWD or laboratory data
Weather → moisture effect → pore-water pressure	Rain intensity and flood hours reduce CBR linearly (floor 0.4); no pore pressure	Assumed	UNVERIFIED. Rainfall is not pore-water pressure; saturation depends on drainage, duration, permeability and groundwater
Moisture → subgrade properties (CBR 6 → 2.5 %)	$E_s = 10\ \mathrm{CBR}\cdot f_w$	Assumed	UNVERIFIED. A soaked/unsoaked CBR pair from tests would be the minimum calibration
Traffic speed / congestion → axle load (virtual WIM)	Not implemented; traffic volume × heavy share × illustrative spectrum × 3 axles	Inferred, labelled	Congestion is correctly **not** treated as truck weight (`evidence.ts` note). The blueprint's speed-to-ODOL rule has no validation and should not be implemented without WIM data
Road age → ageing	Not implemented	—	Requires installation-year data and a mix-specific ageing model
Traffic loading → structural response	Engine EQ-4/5	Calculated (inconsistent with MLET)	FAILED (F-03)
Structural response → predicted deterioration	Not linked: life uses load factor and rain only	Assumed	FAILED (F-07)
Visible pothole → future structural failure	Pothole horizon rule	Assumed	Not a valid inference; potholes have several mechanisms (identifiability limitation)

Undocumented constants in the chain: 0.004 per mm/h, 0.015 per flood hour, floor 0.4; 1.2 % per °C, floor 0.5; the +12 °C air-to-surface assumption (`assessment.ts:86`); 3 axles per heavy vehicle; the illustrative axle spectrum; ±35 % traffic spread. The new layer propagates provenance labels and caps confidence at LOW,

which is appropriate; it does not propagate numerical uncertainty through the engine.

Identifiability. From a photo, a rainfall value and a traffic count, the system cannot uniquely identify whether damage results from overload, moisture, a thin or weak structure, poor mix quality or poor drainage; several mechanisms produce the same surface distress. Any feature that attributes a cause — the "PINN contribution" percentages (F-20) or the planned sub-standard material alert — must present alternatives and treat its output as a hypothesis for investigation.

Chapter 7 — RUL, PCI, and Material Recommendation Audit

$$\Delta_{PCI} = \min\left(35, \frac{2.5LF}{f_w}\right), \quad PCI' = \max\left(10, PCI_0 - 0.8\Delta_{PCI}\right)$$
$$RUL = \max\left(0.3, \frac{PCI'}{1.2\Delta_{PCI}}\right), \quad H_{pothole} = \max\left(3, \frac{1800 f_w}{LF}\right) \qquad \text{(EQ-10)}$$

Engine deterioration rules (AS-IMPLEMENTED, pinnEngine.ts:92-112).

RUL and pothole horizon. Not a time-dependent deterioration process: there is no traffic accumulation, no damage summation and no calibration. Neither quantity depends on asphalt thickness, modulus, CBR or temperature (verified at 18 t across the full range: PCI′ 59, RUL 1.9 years, horizon 171 days in every case). The three time-to-failure quantities contradict each other (1.9 years of life, 171 days to pothole, PCI at the floor of 10 within two years). At zero load RUL and horizon are infinite.

PCI. PCI (as defined by distress-survey standards such as ASTM D6433) is a 0–100 index computed from observed distress types, severities and extents. The engine instead subtracts a load-and-rain term from a stored value, with an undefined horizon ("predicted PCI"), and the five-year "preventive" curve rises 2.5 points a year regardless of inputs and starts 5 points *below* the no-action curve. The scale is valid; the definition and dynamics are not documented FAILED.

Material and overlay recommendation. Overlay thickness is $\max\left(4, 0.22\left(75 - PCI'\right)\right)$ cm when PCI′ < 70 (EQ-11), independent of strain, structure, traffic and material; the type is chosen by thickness bands and one band reads "AC-BC + AC-WC Poly-Modified Asphalt", conflating products. The blueprint's cross-simulation (AC-WC vs PMA under local rain and load, BAB 4.3) and its "PMA ≥ 8.5 cm" directive are not implemented; "8.5 cm" appears only as a UI constant. Recommendations must not be presented as construction-ready designs: an overlay design needs measured deflections or a structural evaluation, design traffic, material properties and a design method (F-18).

Structural anomaly alert. Not implemented (F-26). If built, it must be an investigation trigger (e.g. "response inconsistent with recorded age, traffic and rainfall; core sampling recommended"), never evidence of contractor misconduct; the blueprint's own wording rule (BAB 4.4) is consistent with this.

Predictive accuracy. No accuracy can be claimed: no independent reference observations exist (two dated PCI surveys per segment would be the minimum to begin calibrating a deterioration model).

Chapter 8 — Data Integrity and Validation Dataset Audit

8.1 Computer vision data

Table 8.1 — CV data inventory

Item	Availability	Source / provenance	Status	Training suitability	Validation suitability
Road-damage images	None in repository; sample photos are Unsplash URLs	Unsplash (stock), not annotated	Illustrative	No	No
Labels and severities	Constants per class (`cvDamageClasses.ts`)	Authored	Synthetic	No	No
External notebook (YOLOv8n-seg, Roboflow pothole set, CC BY 4.0, 720/60 images, single class)	Outside repository (`FILES/Docs/…ipynb`, audited in `docs/AI.md`)	Third-party Kaggle run; weights not included	Candidate only	Possible starting point for the pothole class	Not for Indonesian roads without local data
Train/val/test split, leakage checks, anti-spoofing data, class balance	None	—	NOT IMPLEMENTED	—	—

8.2 Pavement-engineering and PINN data

Table 8.2 — Pavement data inventory

Data	Availability	Provenance	Authentic or synthetic	Units / completeness	Suitable for training	Suitable for validation
FWD deflection basins	None	—	—	—	—	—
FEM / MLET reference solutions	None in repository (audit reference generated in `evidence/`, not committed as data)	Audit	Computed	Two/three-layer, on-axis only	Only as physics pre-training; never as sole validation	Only for numerical verification
Layer geometry and materials	Segment defaults (thickness, CBR) + reference baseline (base 20 cm, sub-base 25 cm, modulus 1800–3200 MPa)	Hand-authored; `karawangPilot.ts` header: SYNTHETIC	Synthetic	No ν, no base/sub-base moduli	No	No
Elastic modulus / Poisson ratio	Ranges only; no ν	Authored	Synthetic	Incomplete	No	No
CBR / subgrade	Per-segment defaults (3–6.2 %)	Authored	Synthetic	No soaked/unsoaked pairs	No	No
Traffic / ESAL / WIM	Volumes and heavy shares per segment; illustrative spectrum; legacy "WIM stations" (13)	Authored; pipeline modal uses `Math.random`	Synthetic	No axle spectra	No	No

Data	Availability	Provenance	Authentic or synthetic	Units / completeness	Suitable for training	Suitable for validation
Weather / rainfall / drainage	Deterministic demo values	Hash of segment id	Synthetic	No drainage data	No	No
Longitudinal condition, maintenance history	One "survey date" (DATA REQUIRED) per pilot segment; illustrative histories in legacy views	Authored	Synthetic	No time series	No	No

Synthetic is not ground truth. All segment attributes are synthetic. If MLET or FEM solutions are later generated to train a surrogate, the same generator must not be used as the sole independent validation, because agreement would then measure only the surrogate's fit to its own generator. Independent validation needs field measurements (FWD basins with layer data from cores or as-built records, strain-gauge or instrumented-section data, dated PCI surveys).

Missing data and plausible sources (not verified as accessible). Bina Marga / BBPJN FWD and IRMS survey records for the pilot corridor; as-built drawings and core logs; WIM records (Kemenhub/BPTD or toll operators); BMKG rainfall station series; laboratory soaked/unsoaked CBR; local road-damage photographs with annotation protocol and consent. The audit has no evidence of access to any of these; access requests are a remediation task (R-P2-04). Licensing: Unsplash images are free to use under the Unsplash licence but carry no annotations; the Roboflow dataset is CC BY 4.0 (attribution required).

Chapter 9 — Numerical Verification and Test Audit

The existing suite (226 tests, all passing; Table C.3) is strong for software behaviour and weak for science: 144 unit and 49 integration tests cover workflow, RBAC, API contracts, maps and module boundaries; 33 physics-validation tests prove that the code equals its documented formulas, pin 76 golden scenarios at zero tolerance and characterise threshold reachability. Their own header states that they are "not evidence of physical accuracy". No test compares the engine with an analytical or trusted reference, with FEM or with FWD data; none tests generalisation to unseen segments; there is no CV test and no browser end-to-end test.

Table 9.1 — Coverage of verification objectives

Objective	Existing tests	This audit	Status
Correctness of individual equations (code = formula)	yes (400 random inputs, exact)	re-derived EQ-4..11	VERIFIED
Dimensional consistency	no	analysed (4.3, B.8)	FAILED
Boundary-condition satisfaction	n/a (no BVP in engine)	reference solver: surface traction, interface continuity	engine NOT IMPLEMENTED; reference VERIFIED
Physical limiting behaviour	partial (linearity, monotonicity)	similarity, Boussinesq limit	engine FAILED
Numerical stability	finite outputs in documented ranges	non-finite outside (Table B.7)	PARTIALLY VERIFIED
Comparison with analytical/trusted reference	no	MLET reference, 144 cases	FAILED
Comparison with independent FEM / FWD	no	no data	BLOCKED
Generalisation to unseen segments / loads	no	—	NOT VERIFIED
Reproducibility	golden fixture	non-deterministic telemetry found	PARTIALLY VERIFIED
End-to-end application correctness	HTTP golden path	—	PARTIALLY VERIFIED (no browser automation)

Executed commands, exit codes and limitations are listed in Appendix C (T-01 to T-10); deliberately skipped operations in Table C.2. A passing software test is not proof of scientific validity, and a reference that satisfies its own boundary-value problem is not proof that the model, parameters or predictions are correct for real roads: empirical accuracy remains NOT VERIFIED.

Chapter 10 — Intel OpenVINO and Edge Hardware Audit

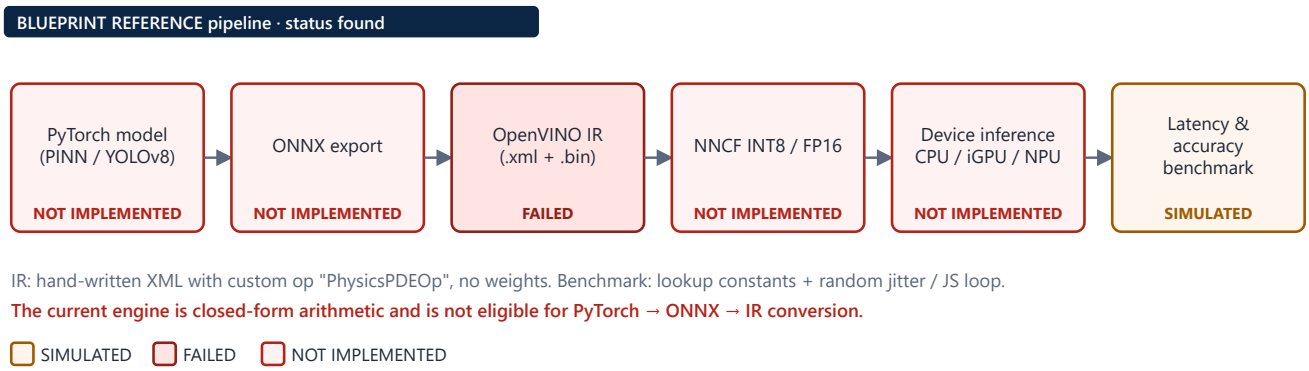


Figure 10.1 — Intended deployment pipeline (BLUEPRINT REFERENCE) with the status of each stage found in the repository.

Table 10.1 — Deployment pipeline status

Stage	Expected artifact	Found	Status
PyTorch model	`.pt` / training code	none; no Python	NOT IMPLEMENTED
ONNX export	`.onnx`	none	NOT IMPLEMENTED
OpenVINO IR	`.xml` + `.bin`	hand-written XML with custom op `PhysicsPDEOp`, no `.bin` (`openvinoExportService.ts:15-42`)	FAILED as an artifact (F-13)
NNCF quantisation	config, calibration set, INT8/FP16 IR	JSON template claiming `nncf_quantization`	NOT IMPLEMENTED
Runtime usage	`openvino` package, `core.compile_model`	none (code samples inside a JSON string only)	NOT IMPLEMENTED
Device availability (CPU/iGPU/NPU)	`available_devices` query	browser heuristic defaulting to "Intel Core Ultra with NPU" (`deviceDetector.ts:14-15,46`)	FAILED (F-12)
Device selection and fallback	AUTO/MULTI policies	UI selector feeding lookup constants	SIMULATED
Precision support	FP32/FP16/INT8/INT4 runs	constant multipliers 1.4/1.0/0.8	SIMULATED
Quantisation deviation	measured output deltas	constants 99.4/99.8/100 %	SIMULATED
Latency (cold/warm), batch, memory, utilisation	measured distributions	constants + random jitter; JS `tanh` loop	SIMULATED (F-02)
< 10 ms target	measured p95 on target device	not measured; "8.4 ms" hard-coded from the blueprint	NOT VERIFIED

Eligibility for the OpenVINO workflow. The current engine is ~40 floating-point operations of closed-form arithmetic. It is not a neural network and cannot be exported through PyTorch → ONNX → IR; it also needs no accelerator (its evaluation time in JavaScript is far below a millisecond by inspection; not timed by the audit). OpenVINO becomes relevant only for (a) a trained CV model, (b) a trained PINN or MLET surrogate, or (c) a

local SLM. For (b), the benchmark must also report numerical deviation of the accelerated, quantised model from its FP32 version **and** from the MLET reference, since quantisation error in strain directly changes fatigue life through the steep transfer functions (EQ-16).

No benchmark number in the application is a hardware measurement. The protocol required before any latency claim is in Appendix F; the benchmark register (Table C.4) is empty by necessity.

Chapter 11 — End-to-End Architecture and Operational Integrity

The intended flow — Public Reporter → Operations Manager → Field Engineer → Physics and Rule Engine → Policy Maker → Repair Execution and Verification — is the strongest part of the implementation.

Table 11.1 — Operational integrity checks

Step	Implementation	Persisted / authorised	Status
Report creation and validation	server identity, photo magic bytes ≤ 5 MB, coordinates plausibility, duplicate hint (30 m / 7 days)	yes / server-side	VERIFIED (tests)
Coordinate and segment association	nearest-segment logic; pilot bounding box approximate (G-01)	yes	PARTIALLY VERIFIED
Evidence image storage	inside the report JSON (≤ 5 MB); no object storage; no EXIF stripping	yes (file store)	PARTIALLY VERIFIED
Weather and traffic provenance	DEMO / SYNTHETIC / INFERRED labels attached	yes	VERIFIED (labels); data NOT IMPLEMENTED
Physics-engine invocation	`assessSegment` on demand; snapshot stored with the recommendation	yes	VERIFIED (software); physics FAILED
Error propagation	400/422/409/503 on workflow and intelligence routes; legacy `/api/simulate` returns 500 or non-finite values	—	PARTIALLY VERIFIED
Recommendation generation	engineer-entered treatment + engine snapshot	yes	VERIFIED
Shared persistence	local atomic JSON store; Supabase not applied	single instance only	BLOCKED
Real-time updates	none (no push channel)	—	NOT IMPLEMENTED
RBAC enforcement	domain + route guards; recommender ≠ approver	yes	VERIFIED (demo identity)
Approval and assignment transitions	legal-transition table; stale-version rejection; serialised writes	yes	VERIFIED
Audit trail	every transition with actor, time, reason; official-only access	yes	VERIFIED
Report generation	client-side jsPDF "official" report	not persisted	PARTIALLY VERIFIED (F-19)
Final repair verification	engineer verifies after work-order completion; OM closes	yes	VERIFIED

UI-only versus backend. The Road detail, What-If, workflow queues and Road Story consume backend engine outputs (rule-based, labelled). The policy dashboards, WIM/ODOL enforcement views, OpenVINO diagnostics, the WIM training modal, many map popups and the classic layout consume constants, random values or a client-side engine call, and several of them carry no status label (F-23). The "3D cross-section" is a 2D drawing whose base layer (15 cm, 300 MPa) is not used by the engine (F-29). No production system was touched and no work order left the application; the internal work order is explicitly not an SPK.

Chapter 12 — Findings, Severity, and Remediation

Thirty-one findings were raised: 5 P0, 15 P1, 8 P2 and 3 P3, together with eight verified strengths. The complete register with all twelve fields per P0/P1 finding is Appendix A; Table 12.1 lists the P0 and P1 findings.

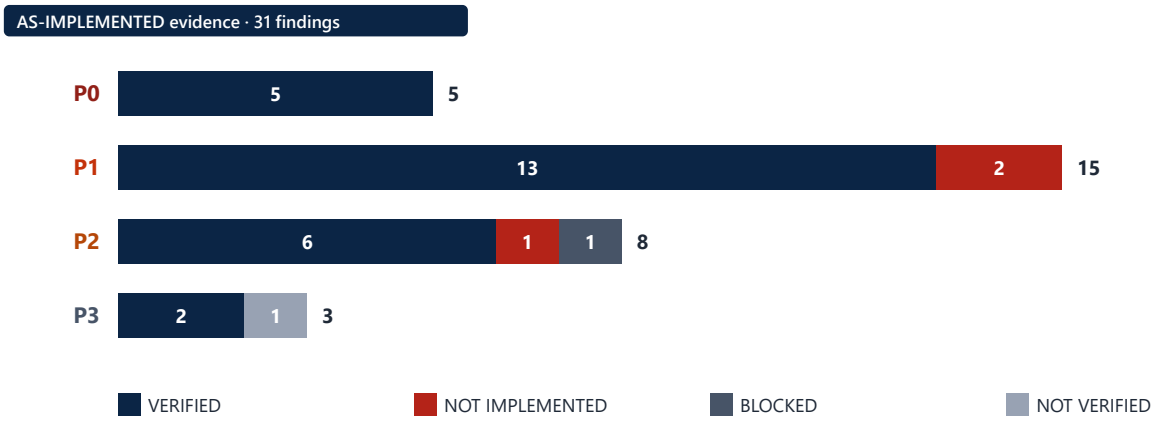


Figure 12.1 — Findings by severity and evidence class. AS-IMPLEMENTED evidence.

Table 12.1 — Critical and high findings

Finding ID	Severity	Component	Evidence	Impact	Required action	Acceptance criteria
F-01	P0	Engine identity	VERIFIED	False PINN claim	Relabel or build a real model	No unsupported "PINN/neural" text; or model artifact + metrics
F-02	P0	Telemetry	VERIFIED	Fabricated AI and Intel figures	Remove random/constant measurements	No `Math.random` in telemetry; figures trace to measurement records
F-03	P0	Strain model	VERIFIED	Unsafe "safe" verdicts	Verified MLET computation	≤ 15 % vs verified reference; similarity exponent −1
F-04	P0	WIM training modal	VERIFIED	Fabricated official data and training	Remove or mock-up label	No epochs/losses/official wording without real run
F-05	P0	CV display	VERIFIED	Invented model confidence	Remove model name/confidence	Confidence only from versioned REAL model
F-06	P1	Thresholds	VERIFIED	Untraceable verdicts	One cited criterion per mode	Threshold register sourced; single constant module
F-07	P1	RUL/PCI	VERIFIED	Baseless life figures	Documented deterioration model	Structural inputs affect life; one failure definition
F-08	P1	Standard axle	VERIFIED	2.255× inconsistency	Single cited axle	Modules agree (test)
F-09	P1	Load model / units	VERIFIED	Non-physical load scaling	Full load contract	Units and dimensions documented
F-10	P1	Moisture/temperature	VERIFIED	Uncalibrated effects	Cite or calibrate	Each relation sourced with validity range
F-11	P1	Legacy API	VERIFIED	Non-finite outputs	Input validation	400/422 for invalid input; no non-finite output
F-12	P1	Device detector	VERIFIED	False NPU claim	Honest detection	NPU only from device query

Finding ID	Severity	Component	Evidence	Impact	Required action	Acceptance criteria
F-13	P1	IR export	VERIFIED	Fake model artifact	Remove or template label	Real IR only from converted model
F-14	P1	OpenVINO pipeline	NOT IMPLEMENTED	Edge claims unsupported	Build after model exists	Reproducible benchmark register
F-15	P1	Blueprint vs spec	VERIFIED	Conflicting requirements	Governance decision	No unexplained DIVERGENT rows
F-16	P1	Data	VERIFIED	No validation basis	Data acquisition	Inventory with provenance; no unsupported badges
F-17	P1	Anti-spoofing	NOT IMPLEMENTED	Spam/spoof risk	Implement and measure	FAR/FRR on held-out set
F-18	P1	Recommender	VERIFIED	Non-design presented as design	Design-method based, advisory	Inputs listed; "not for construction"
F-19	P1	PDF export	VERIFIED	Unlabelled "official" output	Status on every page	Disclaimer; no unauthorised institution names
F-20	P1	Attribution %	VERIFIED	Random causal claims	Remove	No attribution without decomposition

Chapter 13 — Phased Remediation Roadmap

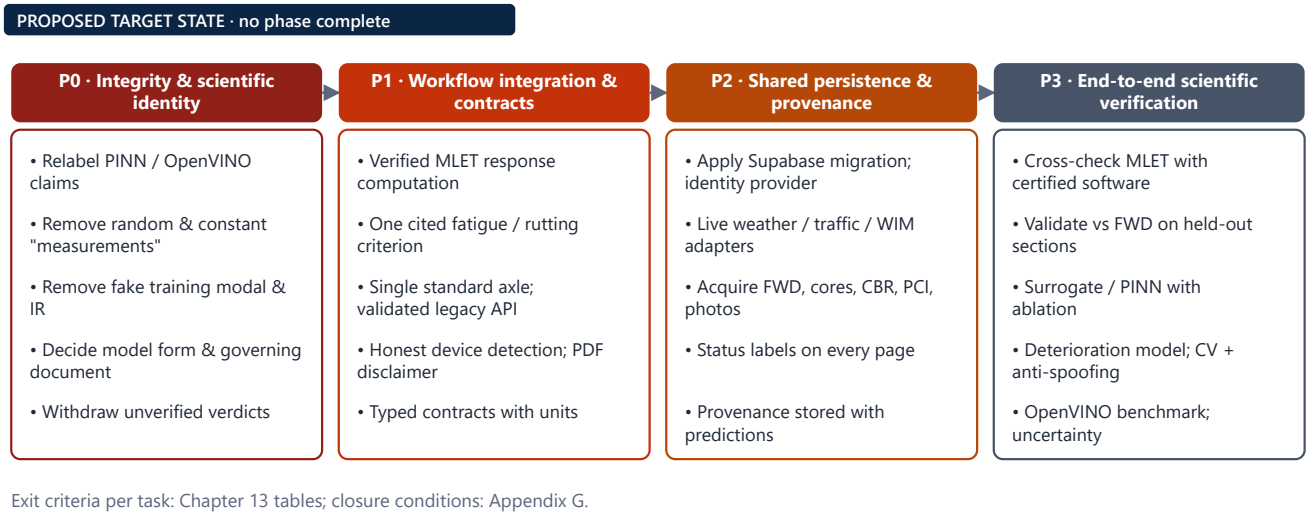


Figure 13.1 — Phased remediation roadmap. PROPOSED TARGET STATE; no phase is complete.

No remediation was implemented during this audit, and no phase is complete until its exit criteria are demonstrated.

Table 13.1 — Phase P0: integrity and scientific identity

Task ID	Priority	Prerequisite	Required change	Acceptance test	Expected evidence	Exit criteria
R-P0-01	P0	—	Replace "PINN / neural / Burmister solver / OpenVINO inference" wording on legacy screens, README and PDF with the engine's actual status	grep test over `src/`, README, PDF template	CI test listing allowed phrases	F-01 closed
R-P0-02	P0	—	Remove `Math.random` and constants from telemetry, loss, benchmark, CV confidence and attribution displays; show "not measured"	unit test: identical inputs → identical payload; grep for `Math.random` in listed files	test output	F-02, F-05, F-20, F-21 closed
R-P0-03	P0	—	Remove or relabel the WIM scraping/training modal and IR export	manual review + route test	screenshots, test	F-04, F-13 closed
R-P0-04	P0	—	Decide model form: (a) MLET screening engine, (b) MLET-trained surrogate, (c) true PINN; record in DECISION_LOG	decision record exists	DECISION_LOG entry	decision approved by team and mentor
R-P0-05	P0	—	Decide governing document (blueprint vs spec v2.2): pilot corridor, stack, 3D, SLM	traceability matrix re-run	updated Appendix D	no unexplained DIVERGENT rows
R-P0-06	P0	R-P0-01	Withdraw engineering verdict text ("STABIL… aman", "MELEBIHI AMBANG") until F-03/F-06 close	grep test	test	no verdicts from unverified responses

Table 13.2 — Phase P1: workflow integration and contracts

Task ID	Priority	Prerequisite	Required change	Acceptance test	Expected evidence	Exit criteria
R-P1-01	P1	R-P0-04	Implement MLET response computation (or adopt a certified library) with full layer inputs, ν, wheel load, pressure, configuration	comparison test vs verified reference (≤ 15 %); Boussinesq limit; similarity	CI test with stored reference grid	F-03, F-09 closed
R-P1-02	P1	R-P1-01	One cited fatigue and one cited rutting criterion with design traffic; single constants module	consistency test across engine, UI, PDF	threshold register with sources	F-06 closed
R-P1-03	P1	—	Single standard axle constant with citation	cross-module test	test	F-08 closed
R-P1-04	P1	—	Validate `/api/simulate` (or retire it in favour of intelligence routes)	contract tests for invalid classes	test output	F-11 closed
R-P1-05	P1	—	Honest device detection; unknown = unknown	unit test with no WebGL info	test	F-12 closed
R-P1-06	P1	R-P0-01	PDF status block and disclaimer on every page; remove institution names	PDF snapshot test	sample PDF	F-19 closed
R-P1-07	P1	—	Typed input/output contract with units for every engine call	schema test	contract doc	contracts versioned

Table 13.3 — Phase P2: shared persistence and provenance

Task ID	Priority	Prerequisite	Required change	Acceptance test	Expected evidence	Exit criteria
R-P2-01	P2	credentials	Apply Supabase migration; implement `CaseStore` on PostgreSQL; RLS tests	integration tests against database	CI logs	F-24 closed
R-P2-02	P2	credentials	Identity provider; retire demo accounts in production	auth tests	config record	REQ-AUTH-002 PASS
R-P2-03	P2	API keys	Live weather adapter; traffic counts; WIM adapter when data exists	provider contract tests	provenance records	F-25 closed or labelled
R-P2-04	P2	institutional contact	Data acquisition plan executed: FWD, cores, CBR, WIM, PCI surveys, local photos	data inventory review	Appendix E updated with provenance and licences	F-16 closed
R-P2-05	P2	R-P0-02	Status label on every page showing model or telemetry output	lint-style UI test	test	F-23 closed
R-P2-06	P2	R-P1-01	Persist model version, inputs, units and provenance with every prediction	schema test	stored records	provenance complete

Table 13.4 — Phase P3: end-to-end scientific verification

Task ID	Priority	Prerequisite	Required change	Acceptance test	Expected evidence	Exit criteria
R-P3-01	P3	R-P1-01	Cross-check the MLET implementation against a certified program (KENLAYER, WESLEA, BISAR or PyMastic)	agreement within stated tolerance on ≥ 20 structures	comparison table	reference accepted
R-P3-02	P3	R-P2-04	Validate against FWD basins (back-calculation) on held-out sections	RMSE, MAE, bias per sensor offset	validation report	blueprint BAB 10.1 metrics reported honestly
R-P3-03	P3	R-P0-04, R-P2-04	If (b)/(c): build surrogate or PINN with physics residual, data loss, documented split; ablation (data-only vs data+physics)	held-out metrics; out-of-range behaviour	training logs, checkpoints, model card	F-01 closed in the strong sense
R-P3-04	P3	R-P1-02, R-P2-04	Deterioration and RUL model from transfer functions and/or calibrated PCI curves with traffic growth	calibration and validation on dated surveys	model report	F-07, F-18 closed
R-P3-05	P3	R-P2-04	CV model trained and evaluated on local annotated photos; anti-spoofing with FAR/FRR	mAP, per-class metrics, leakage checks	model card	F-05, F-17 closed
R-P3-06	P3	R-P3-03 or R-P3-05	OpenVINO conversion, NNCF quantisation and benchmark on named Intel hardware	Appendix F protocol	benchmark register filled	F-14 closed
R-P3-07	P3	all	Uncertainty quantification and limitation statement for every output	review	model cards	claims register updated

Chapter 14 — Final Scientific Readiness Assessment

Table 14.1 — Readiness by dimension

Dimension	Current status	Supporting evidence	Critical limitation	Required next step	Public claim now?
Genuine PINN implementation	NOT IMPLEMENTED	Table 3.1	No network, residual, data or training	R-P0-04, R-P3-03	No
Mathematical correctness (code = formula)	VERIFIED	existing tests; EQ-4..11	formulas themselves unvalidated	—	Only as "formula engine"
Physics consistency	FAILED	Tables B.3–B.5; similarity	1–2 orders below MLET	R-P1-01	No
Strain and failure-threshold traceability	FAILED	Table B.6	no sources; inconsistent	R-P1-02	No
Training-data readiness	NOT IMPLEMENTED	Tables 8.1–8.2	no empirical data	R-P2-04	No
Independent numerical verification	PARTIALLY VERIFIED	audit reference (closed-form verified)	not yet cross-checked with certified software; not in CI	R-P3-01, R-P1-01	Engine: no
Empirical validation	NOT VERIFIED	—	no FWD/PCI data	R-P3-02	No
RUL and PCI credibility	FAILED	EQ-10, Table B.7	decoupled, inconsistent	R-P3-04	No
Application integration	VERIFIED (workflow) / PARTIALLY VERIFIED (AI stages)	226 tests; Table 11.1	demo identity; local store	R-P2-01, R-P2-02	Yes, as a workflow prototype
OpenVINO readiness	NOT IMPLEMENTED	Table 10.1	no model, no runtime	R-P3-06	No
Reproducibility	PARTIALLY VERIFIED	golden fixture; non-deterministic telemetry	random payload fields	R-P0-02	Partially
Competition-claim readiness	FAIL	Chapters 3, 4, 10	five open P0	Phase P0	Only claims in list 1

14.1 Claims supported by current evidence

1. A working multi-stakeholder road-damage reporting and maintenance workflow with four roles, server-enforced permissions, separation of duties, persistence and a complete audit trail (demo identity, local storage).
2. Anonymous citizen reporting with server-issued identity and validated photo uploads.
3. A deterministic, rule-based pavement screening calculation whose outputs are labelled RULE_BASED with LOW confidence on the newer screens, with input provenance and out-of-range flags.
4. A correct ESAL / fourth-power load-equivalency calculation (with the 8.16 t standard axle).
5. Persisted What-If scenario comparisons labelled as simulations.
6. An honest internal validation-gap analysis and this independent audit, including a closed-form-verified layered-elastic reference implementation.

14.2 Claims requiring further verification

1. That the formula engine can serve as a screening tool after its responses are replaced or recalibrated against a verified MLET computation (R-P1-01).
2. That fixed strain limits or transfer functions from a named standard apply to the pilot corridor (R-P1-02).
3. Camera and GPS reporting on real phones within the blueprint's 15-second target.
4. Weather, traffic and WIM integration once data sources are connected.
5. Any CV accuracy figure, after a model is trained on local data.
6. Any latency, throughput or power figure, after a model exists and is benchmarked on named Intel hardware.

14.3 Claims that must not yet be made

1. That PAVEMENT-PINN contains, trains or runs a Physics-Informed Neural Network.
2. That predictions obey Burmister / multilayer elastic theory or are "equivalent to FEM".
3. That any physics loss, data loss, training epoch or convergence value has been observed.
4. That inference runs on Intel OpenVINO, on an NPU or iGPU, in INT8/FP16, or in under 10 ms (including "8.4 ms").
5. That YOLOv8 or any model classifies damage, with any confidence value, or that photos are screened for spoofing.
6. That remaining useful life, pothole horizon, PCI projections, overlay thickness, material choice, economic loss or budget savings are predictions of a validated model.
7. That data are scraped from, validated by or endorsed by Bina Marga, PUPR, Kemenhub, BMKG or Intel.
8. That a road segment is structurally "safe" or "exceeds the permitted limit" on the basis of the current engine.
9. That the system has been validated with FWD, WIM or field data.

Appendix A — Findings Register (Artifact B)

Document PAVEMENT-PINN-AUDIT-001 · Version 1.0 · Audit date 2026-10-14 · Commit `4a9b4fc` (branch `feature/m1-report-ownership`, working tree with uncommitted demo-authentication changes from another session; see Document Control).

Severity: P0 critical, P1 high, P2 medium, P3 low. Evidence classes: VERIFIED, PARTIALLY VERIFIED, NOT IMPLEMENTED, NOT VERIFIED, FAILED, BLOCKED. Line numbers refer to the working tree at the audit date. "Reference solver" means the audit's independent multilayer linear-elastic solver (`evidence/mlet.mjs`), itself verified only against closed-form limits (Appendix B).

A.1 Summary register

Table A.1 — Findings register summary (31 findings: 5 P0, 15 P1, 8 P2, 3 P3)

ID	Sev.	Title	Component	Evidence	Acceptance criterion (abridged)
F-01	P0	Deterministic closed-form engine presented as a trained PINN	`src/engines/pinn/core/pinnEngine.ts`; UI titles; README	VERIFIED	Every user-facing "PINN"/"neural" claim either backed by a trained model artifact or replaced by "rule-based formula engine"
F-02	P0	Fabricated model and runtime telemetry (losses, latency, power, NPU use)	`pinnEngine.ts:203-240`; `FieldEngineerTools.tsx:563-583`; `Navbar.tsx:225-229`; `PavementPinnDashboard.tsx:186-189`; `pdfGenerator.ts:90`	VERIFIED	No numeric loss, latency, power or utilisation shown unless measured; random generators removed from result payloads
F-03	P0	Engine strains one to two orders of magnitude below layered-elastic theory; "safe" verdicts unsupported	`pinnEngine.ts:58-76,127-132`	VERIFIED	Engine within an agreed tolerance (e.g. ±15 %) of a verified multilayer reference over the stated input domain, or engine withdrawn from engineering verdicts
F-04	P0	Simulated "Bina Marga WIM scraping + PINN training" with synthetic epoch-loss curve	`BinaMargaWimPipelineModal.tsx`; `binaMargaWimScraper.ts:70-155`	VERIFIED	Modal removed, or relabelled as a mock-up inside the modal itself, with no epochs, losses or official-source wording
F-05	P0	"YOLOv8 Edge AI" damage class and confidence are keyword matches and constants	`cvDamageClasses.ts:24-244`; `ReportRoadIssueCard.tsx:1051-1093`	VERIFIED	No model name or confidence shown to citizens unless produced by a versioned model run
F-06	P1	Strain limits 180/200/250/350 με untraceable, mutually inconsistent, and partly hard-coded as "exceeded"	`pinnEngine.ts:128-132`; `FieldEngineerTools.tsx:193-209,397`; `pdfGenerator.ts:111`	VERIFIED	One cited criterion per quantity, tied to load repetitions, used consistently in engine, UI and PDF
F-07	P1	RUL, pothole horizon and PCI projection are heuristics decoupled from structural response and mutually inconsistent	`pinnEngine.ts:90-112,173-186`	VERIFIED	Single documented deterioration model; thickness, modulus and CBR affect life; one time-to-failure definition
F-08	P1	Two different fourth-power reference axles (10 t vs 8.16 t) in one application	`pinnEngine.ts:54`; `esal.ts:12`	VERIFIED	One standard axle used throughout, with cited source
F-09	P1	Load model lacks contact pressure, radius, wheel configuration and Poisson ratios; dimensionally inhomogeneous fits	`pinnEngine.ts:56-80`; `types/simulation.ts`	VERIFIED	Load defined as wheel load + contact pressure (or radius) + configuration; all inputs with units; dimension check documented
F-10	P1	Moisture, temperature and subgrade relations are undocumented linear heuristics; no pore-water pressure anywhere	`pinnEngine.ts:35-50`	VERIFIED	Each relation cited or calibrated, with stated validity range and uncertainty
F-11	P1	Legacy `/api/simulate` accepts any input; returns non-finite, negative or coerced results	`simulationController.ts`; `pinnEngine.ts`	VERIFIED (function level)	Schema validation with ranges; 400/422 for invalid input; no non-finite value in any response

ID	Sev.	Title	Component	Evidence	Acceptance criterion (abridged)
F-12	P1	Hardware detector defaults to "Intel Core Ultra with NPU" and infers an NPU from core count	`deviceDetector.ts:14-15,46`	VERIFIED	Unknown hardware reported as unknown; NPU only from a positive device query
F-13	P1	"OpenVINO IR export" is a hand-written XML with a custom op and no weights	`openvinoExportService.ts:15-42,53`	VERIFIED	Endpoint removed or relabelled as a template; real IR only from a converted trained model
F-14	P1	No OpenVINO runtime, model, quantisation or benchmark; "< 10 ms" and "8.4 ms" unsupported	repository-wide; `simulatedBenchmark.ts`	NOT IMPLEMENTED	Measured latency distribution and accuracy deltas on named hardware, reproducible script
F-15	P1	Blueprint vs implementation source-of-truth conflict (pilot corridor, stack, 3D twin, on-device SLM)	`karawangPilot.ts`; `screening.ts:40,83`; `package.json`; `narrativeService.ts:9`	VERIFIED	Written decision on which document governs; blueprint or code updated so they agree
F-16	P1	No empirical dataset in the repository; contradictory provenance badges ("FWD Validated", WIM stations)	`indonesianRoads.ts`; `LeafletMap.tsx:757,988`; `karawangPilot.ts`	VERIFIED	Data inventory with source, licence, status per dataset; badges removed where no data exists
F-17	P1	Anti-spoofing (photo authenticity) not implemented	repository-wide	NOT IMPLEMENTED	Implemented check with measured false-accept/false-reject rates on a held-out set
F-18	P1	Overlay thickness and material type are PCI look-ups, not structural designs	`pinnEngine.ts:115-125`	VERIFIED	Recommendation derived from a design method with traffic, structure and material constraints, labelled advisory
F-19	P1	Exported "official" PDF carries simulated telemetry and heuristic verdicts without a disclaimer	`pdfGenerator.ts:78-163`	VERIFIED	PDF states engine status, data status and "not for construction" on every page
F-20	P1	Random "PINN contribution" percentages attributed to damage causes	`CommunityReportsDashboard.tsx:259-261`; `LeafletMap.tsx:360-361`	VERIFIED	No attribution percentage without a decomposition method; random calls removed
F-21	P2	Non-deterministic result payloads (same input, different output)	`pinnEngine.ts:217,238-239,284`	VERIFIED	Identical inputs give identical outputs (except explicit ids/timestamps)
F-22	P2	Economic loss, savings, Zero-ODOL impact and life-extension figures are invented constants	`pinnEngine.ts:188-201`; policy views	VERIFIED	Each figure tied to a cost model and data source, or removed
F-23	P2	Simulation/illustration labels missing on several pages and the classic layout	pages listed in A.2	VERIFIED	Every page showing engine, CV or telemetry output carries an adjacent status label
F-24	P2	Shared persistence not connected; schema diverges from blueprint	`supabase/migrations/…sql`; `fileCaseStore.ts`	BLOCKED	Migration applied to a project; integration tests against it; schema decision recorded
F-25	P2	Weather and traffic inputs are deterministic demo providers	`evidence.ts:56-153`	VERIFIED	Live provider with provenance, or demo label retained everywhere it appears

ID	Sev.	Title	Component	Evidence	Acceptance criterion (abridged)
F-26	P2	Ageing curve, sub-standard material alert and AC-WC/PMA cross-simulation not implemented	repository-wide	NOT IMPLEMENTED	Implemented with documented logic; alert framed as investigation trigger
F-27	P2	Test suite has no independent-reference, empirical, CV or browser E2E tests	`tests/`	VERIFIED	Reference-comparison test in CI; E2E golden path automated
F-28	P2	`assessSegment` clamps out-of-range inputs and reports outputs for the clamped inputs	`assessment.ts:120-141`	VERIFIED	Out-of-range request refused, or outputs explicitly tagged as computed for the clamped values
F-29	P3	2D cross-section shows a base layer (15 cm, 300 MPa) that the engine never uses; total thickness label inconsistent	`SimulationPanel.tsx:468,495`	VERIFIED	Drawing reflects engine inputs exactly
F-30	P3	Stale or contradictory documentation and labels	`README.md:1,262`; `SimulationPanel.tsx:704`; `package.json`	VERIFIED	README test count, title and model names match the code
F-31	P3	Export filename built from an unsanitised query parameter	`openvinoController.ts`	NOT VERIFIED	Parameter whitelisted to known devices/precisions

A.2 Detailed findings

F-01 — Deterministic closed-form engine presented as a trained PINN

Field	Content
Severity / class	P0 · VERIFIED
Affected	`src/engines/pinn/core/pinnEngine.ts` (`runPinnSimulation`, 296 lines; doc comment l.14-18 "Runs the Physics-Informed Neural Network (PINN) simulation"); `src/pages/engine/SimulationPage.tsx:23`; `src/app/layouts/Navbar.tsx:91`; `src/components/simulation-viewer/SimulationPanel.tsx:154,164,452,516`; `README.md:1`
Blueprint requirement	BAB 3 Fase 1 and BAB 5.1: neural network trained with $\mathcal{L}_{total}$ (data, physics, boundary, initial-condition loss), Adam + L-BFGS
Observed	One pure function of 9 scalars. No network definition, no weights, no optimiser, no automatic differentiation, no residual, no dataset, no checkpoint. `package.json` has no ML dependency; no Python in the repository. The newer domain layer labels it honestly (`assessment.ts:17` "deterministic, no trained weights"; `pages/engine/PinnPage.tsx:35-37` "PLANNED"), but legacy screens still present it as a PINN/OpenVINO solver.
Explanation	A PINN is a parameterised function u_θ trained so that a PDE residual evaluated by differentiation of u_θ is small. None of these elements exists; the outputs are power-law expressions with literal coefficients (EQ-4 to EQ-6).
Impact	Competition and stakeholder claims of a "PINN" would be factually false for the current build.
Corrective action	Rename the engine in all user-facing text ("formula-based screening engine"), keep the PINN as roadmap; or build and train a real model (Chapter 13, R-P3-03).
Acceptance criteria	(1) grep of `src/` and README for "Physics-Informed Neural Network", "PINN Solver", "neural" returns only roadmap or labelled-status text; or (2) a versioned model artifact, training script, data split and held-out metrics exist and the UI loads that model.
Dependencies	Product decision (F-15).
Verification	Code inspection; `npm test` suite `pinn-mathematical-audit.test.ts` (code equals closed-form formulas).

F-02 — Fabricated model and runtime telemetry

Field	Content
Severity / class	P0 · VERIFIED
Affected	`pinnEngine.ts:204-239` (latency = per-device constant × precision factor + `Math.random()*0.8-0.4`; power, CPU/NPU utilisation constants; `physicsLossValue = 0.0018 + Math.random()*0.0006`; `dataLossValue = 0.0011 + Math.random()*0.0004`). Displays: `FieldEngineerTools.tsx:563-583` ("Loss Fisika PDE (Burmister)", "Konvergensi Adam Optimizer epoch 1500.", "Kesesuaian dengan sensor regangan WIM & FWD"); `Navbar.tsx:225-229`; `PavementPinnDashboard.tsx:186-189` (hard-coded "Intel NPU • 8.4 ms", comment "From PDF"); `OperationsHomeView.tsx:615`, `OperationsManagerAppView.tsx:480-484` ("6.2 ms"); `OpenVINOModelDiagnosticsModal.tsx:257-277` ("Hasil Benchmark 50-Pass Inference Pada Perangkat Ini", "Akurasi Quantization NNCF" 99.4 %); `runtime/openvino/simulatedBenchmark.ts:29-76`; `pdfGenerator.ts:78,90`
Blueprint requirement	BAB 5.2, BAB 7.2 telemetry badge "Intel OpenVINO INT8 · NPU + iGPU · Latency 8.4 ms" — described in the blueprint as a benchmark result of a model that does not exist yet

Field	Content
Observed	Loss values and latencies change between identical calls (edge test: 34.1 vs 34.2 ms; physics loss 0.00185 vs 0.00227). The diagnostics modal reports a "benchmark on this device" computed from a lookup table blended with a JavaScript `tanh` loop.
Explanation	These are not measurements of any model or device. A loss value implies a training or residual evaluation; none occurs.
Impact	Directly misrepresents AI behaviour and Intel hardware performance — the two claims most visible to competition judges.
Corrective action	Remove random generators from result payloads; show "not measured" until real measurements exist; label the diagnostics modal as a mock-up inside the modal.
Acceptance criteria	No `Math.random` in any engine or telemetry path (`grep -rn "Math.random" src server` returns none in these files); every latency/loss figure in the UI traces to a stored measurement record with device, runtime version and date.
Dependencies	F-14 for real measurements.
Verification	Code inspection; `evidence/edge.mjs` (non-determinism); subagent UI sweep spot-checked at the cited lines.

F-03 — Strain responses inconsistent with layered-elastic theory; unsafe "STABIL" verdicts

Field	Content
Severity / class	P0 · VERIFIED (against the audit reference solver; the solver itself is verified against Boussinesq and Burmister two-layer chart values only)
Affected	`pinnEngine.ts:58-76` (EQ-4 to EQ-6), `pinnEngine.ts:127-132` (engineering notes "STABIL … dalam batas elastis aman")
Blueprint requirement	BAB 5.1: responses governed by Burmister multilayer elastic theory
Observed	Over 144 in-range dry cases (P 10-25 t, h 5-25 cm, E_1 1200-3500 MPa, CBR 3-10), engine/reference ratios: εt median 0.047 (two-layer) and 0.067 (three-layer with 15 cm base), maximum 0.79; εv median 0.019 and 0.025, maximum 0.063. Blueprint default segment (AC 10 cm, 2500 MPa, CBR 6) at 10 t: engine εt = 20 με, εv = 17 με; reference 303-435 με and 793-1174 με. Elastic similarity fails (engine exponents −0.4/−0.75/−0.6 instead of −1); local sensitivities differ in sign pattern (engine εt independent of CBR, εv independent of E_1).
Explanation	Any linear-elastic layered solution scales as $1 / k$ when every modulus is multiplied by k (EQ-14) and couples all layers. The engine's independent power laws cannot satisfy either condition. Under any plausible single-axle loading, the responses are far below elastic predictions, so the engine labels typical structures as safe.
Impact	Engineering verdicts and every downstream quantity that uses the strains are not physically meaningful; a "safe" message can be produced for a structure that elastic theory places well above common fatigue limits.
Corrective action	Replace the response formulas with a verified multilayer elastic computation (or a surrogate trained on one), with full layer inputs; until then remove verdict language.
Acceptance criteria	Automated comparison test: engine vs verified reference on a documented grid, relative error ≤ 15 % (or a tolerance agreed and justified by the team) for εt, εv and w_0; elastic-similarity test passes with exponent −1 ± 0.01.
Dependencies	F-09 (load definition), F-16 (structure data).

Field	Content
Verification	`evidence/compare.mjs`, `evidence/evidence-run.log`; Appendix B.4.

F-04 — Simulated WIM scraping and "PINN training"

Field	Content
Severity / class	**P0** · **VERIFIED**
Affected	`src/features/wim/BinaMargaWimPipelineModal.tsx` (buttons l.155-159 "Jalankan Scraper & PINN Training"; l.245 "...untuk melatih Physics-Informed Neural Network (PINN)"; l.313-331 "Progres Minimisasi Loss Jaringan (100 Epochs)"); `src/services/wim/binaMargaWimScraper.ts:70-115` (28 records from `Math.random`), `:121-155` (losses are fixed multiples of a mean, epoch history = total loss × 4.2/2.8/1.9/1.3/1.0)
Blueprint requirement	BAB 3 Fase 1-2 (training, WIM data)
Observed	No HTTP request is made to `binamarga.pu.go.id`; no training occurs; the "loss curve" is a fixed geometric pattern. The modal offers a download named `BinaMarga_WIM_PINN_Training_Dataset.json` (l.81). Status label exists only on `PinnPage.tsx:45-47,114`, not inside the modal or on the Navbar entry.
Impact	Presents fabricated official data and a fabricated training run. Also implies scraping of a government portal, which the blueprint itself does not specify.
Corrective action	Remove, or replace with a clearly labelled design mock-up without numbers.
Acceptance criteria	Modal contains no epoch counts, losses, record counts or official-source wording unless backed by a real run log.
Verification	Code inspection (spot-checked lines 121-155 and 238-246).

F-05 — CV classification is a keyword match with constant confidence

Field	Content
Severity / class	**P0** · **VERIFIED**
Affected	`src/engines/computer-vision/classification/cvDamageClasses.ts:216-244` (`analyzeImageDamageClass`: substring tests on the image URL/filename such as `'alligator'`, `'retak'`, Unsplash photo ids; default `pothole`), constants `defaultConfidence` 87-95 (l.38-200); display `ReportRoadIssueCard.tsx:1051,1068,1093` ("Model Engine: YOLOv8 Edge AI", "Confidence: 94%")
Blueprint requirement	BAB 3 Fase 3 (YOLOv8 classification with confidence and severity)
Observed	For an uploaded camera photo (a `data:` URL) the function matches substrings against base64 text — almost always the default "Pothole, 94 %". Mitigation that works: the server-side screening treats the classifier as `DEMO` and always routes to human review (`screening.ts:27-34,88-97`; test "the DEMO classifier always routes to human review" passes).
Impact	Citizens see a model name and a confidence that no model produced.
Corrective action	Show "classification not available (demo)" until a versioned model runs server-side.

Field	Content
Acceptance criteria	Confidence shown only when `modelStatus = REAL` with `modelVersion`; mAP and per-class metrics on a held-out local set documented.
Dependencies	F-16 (dataset), F-17.
Verification	Code inspection; unit tests in `tests/unit/workflow.test.ts`.

F-06 — Strain thresholds untraceable and inconsistent

Field	Content
Severity / class	P1 · VERIFIED
Affected	`pinnEngine.ts:128-132` (KRITIS if εt > 250, text cites "batas ijin Bina Marga (200 με)"; WASPADA if > 180); `FieldEngineerTools.tsx:193-209` (`BatasIjinAashto` 200 and 350, fallback values 245 and 380), `:397` (hard-coded "MELEBIHI AMBANG BATAS (+22.5%)"), legends "Batas Ijin AASHTO" (l.391) vs "Batas Ijin Bina Marga" (`FieldEngineerDashboard.tsx:199`) for the same field; `SimulationPanel.tsx:563-571`; `pdfGenerator.ts:111` ("AMBANG LELEH (>200 με)")
Blueprint requirement	BAB 3 Fase 4, BAB 4.3 (failure prediction from εt, εv)
Observed	No document, clause, table or calibration is cited for any value. Engine output exceeds 200 με in 3 of 144 grid points and never exceeds 350 με; the reference solver exceeds them in 77-85 % and 88-93 % of the same points (Appendix B.5). The "+22.5 %" banner is displayed whatever the result. "Ambang leleh" (yield threshold) is not a concept of linear-elastic fatigue design.
Explanation	Mechanistic-empirical criteria relate an allowable strain to a number of load repetitions through a transfer function (EQ-16 form); a single fixed strain presupposes an unstated design traffic.
Impact	Exceedance messages are both unreachable (engine scale) and, where shown, partly hard-coded.
Corrective action	Choose one criterion per failure mode (fatigue: εt at bottom of asphalt; rutting: εv at top of subgrade), cite it, and evaluate it with design traffic.
Acceptance criteria	Threshold register (Appendix B.5) has a source for each value; one constant module used by engine, UI and PDF; test asserts consistency.
Verification	Code inspection; `compare.mjs` reachability counts.

F-07 — RUL, pothole horizon and PCI projection decoupled from structure and mutually inconsistent

Field	Content
Severity / class	P1 · VERIFIED
Affected	`pinnEngine.ts:90-112` (annual drop, predicted PCI, remaining years, pothole horizon), `:173-186` (5-year projection)
Observed	At 18 t (dry): predicted PCI 59, remaining life 1.9 years, pothole horizon 171 days, no-action PCI reaching the floor of 10 in 2028 — for every asphalt thickness 5-25 cm, modulus 1200-3500 MPa, CBR 3-10 and temperature (edge test). "Preventive" PCI in the first year is 5 points below "no action". Zero load gives remaining life and pothole horizon = ∞ (serialised as `null`).

Field	Content
Explanation	PCI (ASTM D6433 sense) is a distress-survey index; a projection needs an observed deterioration model. Here life depends only on load factor and rain, not on any structural response.
Impact	Life and horizon numbers presented to engineers and policy makers have no structural basis and contradict each other.
Corrective action	Define one deterioration model (e.g. damage accumulation from strain-based transfer functions with traffic growth, or a calibrated PCI curve), with time unit and horizon stated.
Acceptance criteria	Life metrics respond monotonically to structural inputs in documented directions; one definition of "failure"; finite outputs for all valid inputs; calibration evidence or explicit ILLUSTRATIVE label.
Verification	`evidence/edge.mjs` output.

F-08 — Inconsistent fourth-power reference axle

Field	Content
Severity / class	P1 · VERIFIED
Affected	`pinnEngine.ts:54` (`(axle/10)^4`, comment "legal single axle load = 10 Ton"); `src/domain/intelligence/esal.ts:12` (`STANDARD_AXLE_TON = 8.16`, 80 kN)
Observed	The same axle has load factors differing by $(10 / 8.16)^4 = 2.255$ between the two modules; UI text "mempercepat kerusakan N× lipat" uses the 10 t base.
Acceptance criteria	Single constant, cited; tests assert both modules agree.

F-09 — Load model and dimensional homogeneity

Field	Content
Severity / class	P1 · VERIFIED
Affected	`pinnEngine.ts:56-80`; `src/types/simulation.ts:6-19`
Observed	Load enters only as axle tonnes; no tyre pressure, contact radius, wheel/axle configuration, Poisson ratio, base or sub-base layer. `stressTopMpa = 0.65 × P/10` makes contact pressure proportional to axle load (physically, contact pressure ≈ tyre pressure and the contact area grows with load). Strains are linear in P (exponent 1) where the reference gives 0.56 (εt) and 0.64 (εv) at the blueprint point. Coefficients 120, 210, 45 carry hidden units (με·cm^1.75/t etc.).
Acceptance criteria	Input contract lists wheel load (kN), contact pressure (kPa) or radius (mm), configuration, layer E (MPa), ν, h (mm); dimensional analysis note in the repository.

F-10 — Undocumented moisture, temperature and subgrade relations; no pore-water pressure

Field	Content
Severity / class	P1 · VERIFIED
Affected	`pinnEngine.ts:37-50`

Field	Content
Observed	CBR multiplied by $\max\left(0.4, 1 - 0.004R - 0.015t_f\right)$ with rain intensity R (mm/h) as a direct proxy for saturation; $E_s = 10$ CBR; asphalt modulus reduced 1.2 %/°C above 25 °C (floor 0.5). No pore-water-pressure quantity exists in code (blueprint BAB 3 Fase 3, BAB 8.2 `pore_water_pressure_kpa`). Blueprint's own mappings (modulus penalty 40-60 % for high-severity alligator cracking, CBR 6 → 2.5 % when saturated) are not implemented, and the newer translation layer deliberately refuses photo → modulus mapping (`assessment.ts:238-257`), which the audit regards as correct.
Explanation	Rain intensity is not soil moisture; saturation depends on drainage, duration, permeability and groundwater. Temperature effects on asphalt modulus are mix-specific and non-linear.
Acceptance criteria	Each relation documented with source or calibration data and validity range; outputs labelled INFERRED with uncertainty.

F-11 — Unvalidated legacy simulation endpoint

Field	Content
Severity / class	P1 · VERIFIED at function level; HTTP path NOT VERIFIED (the user's server on port 3000 was not called)
Affected	`server/controllers/simulationController.ts` (comment: "No input validation is performed"); `server/services/simulationService.ts`
Observed	Direct calls: zero load → remaining life ∞, pothole horizon ∞, Zero-ODOL −∞; negative load → negative strains (−10 με) and a pothole horizon of 28 800 days; thickness 0 or modulus 0 → infinite strain; CBR ≤ 0 → infinite subgrade strain; `NaN` → `NaN` throughout; string `"12"` silently coerced. JSON serialises non-finite values as `null`.
Acceptance criteria	Validation mirroring the intelligence routes (`intelligence.routes.ts:95-110`); contract tests for each invalid class.

F-12 — Hardware detector asserts Intel NPU by default

Field	Content
Severity / class	P1 · VERIFIED
Affected	`src/runtime/intel-hardware/deviceDetector.ts:14-15` (defaults `'Intel Corporation'`, `'Intel® Core™ Ultra Processor with NPU & Intel® Graphics'` when WebGL renderer info is unavailable), `:46` (`hasNpuAccelerator` true for WebNN presence or any Intel renderer with ≥ 12 logical cores)
Acceptance criteria	Unknown hardware reported as unknown; NPU claimed only from an OpenVINO `available_devices` query on the host.

F-13 — Illustrative IR export

Field	Content
Severity / class	P1 · VERIFIED (content); loadability BLOCKED (no OpenVINO runtime available to attempt loading)
Affected	`server/services/openvinoExportService.ts:15-42` (hand-written `<net>` with `type="PhysicsPDEOp" version="custom"`, no `.bin`), `:53` (`openvino_version: "2024.5.0"`)
Observed	The file header comment states the artifacts are illustrative; the UI download does not. A MatMul layer without weights and an unknown custom op cannot form an executable OpenVINO model.
Acceptance criteria	Endpoint returns 404/“not available” until a converted model exists, or files are named and captioned as templates.

F-14 — OpenVINO deployment pipeline absent

Field	Content
Severity / class	P1 · NOT IMPLEMENTED / measurement BLOCKED
Observed	No PyTorch model, ONNX file, IR (.xml/.bin), NNCF config, OpenVINO runtime dependency, Python environment (`docs/GAP_REGISTER.md` G-PY-01; confirmed: `python` not installed on the audit machine) or benchmark script. Because the engine is a closed-form expression, the PyTorch→ONNX→IR path has no subject; a closed-form engine needs no accelerator (it evaluates in microseconds in JavaScript).
Acceptance criteria	Benchmark register (Appendix C.4) filled from a reproducible script: device, OpenVINO version, precision, batch, warm-up, ≥ 1000 timed runs, p50/p95/p99, cold start, peak memory, and max absolute/relative output deviation vs FP32.

F-15 — Blueprint vs implementation conflict

Field	Content
Severity / class	P1 · VERIFIED
Observed	(a) Pilot: blueprint BAB 1.2 names Jalan Raya Narogong (Bekasi–Bogor); the code implements Karawang Pantura Klari–Jatisari (`karawangPilot.ts`, from spec v2.2 per `docs/SOURCE_INVENTORY.md`). "Narogong" appears nowhere in `src/`; a report at approximately (−6.38, 107.00) is flagged `OUT_OF_SCOPE` (`screening.ts:40,83`; edge test). (b) Stack: blueprint Next.js + Supabase + FastAPI/PyTorch/OpenVINO; code Vite + React 19 + Express + JSON file store. (c) Blueprint 3D "Holographic Cross-Section" (Three.js) and isometric map (pitch 45-60°): no Three.js dependency, no map pitch setting; the cross-section is a 2D stack of `div`s. (d) Blueprint on-device SLM (Phi-3/Llama-3.2 INT4 on NPU, "tanpa membocorkan data … ke server pihak ketiga"): code calls a cloud model (`narrativeService.ts:9`, `gemini-3.7-flash`) with a template fallback.
Acceptance criteria	Recorded decision naming the governing document; traceability matrix (Appendix D) re-run with no unexplained DIVERGENT rows.

F-16 — No empirical data; contradictory provenance badges

Field	Content
Severity / class	P1 · VERIFIED
Observed	Repository contains no FWD basins, FEM solutions, CV images/weights, WIM records, cores or longitudinal PCI surveys. Segment attributes are hand-authored (`karawangPilot.ts` header: SYNTHETIC/ILLUSTRATIVE). The legacy dataset declares 13 "Stasiun WIM" records and `hasWimStation: true` (`indonesianRoads.ts:26,77,128…`) while the pilot declares no WIM; map popups show "FWD Validated" (`LeafletMap.tsx:757,988`).
Acceptance criteria	Data inventory (Appendix E) complete; no validation badge without a validation record.

F-17 — Anti-spoofing absent

Field	Content
Severity / class	P1 · NOT IMPLEMENTED
Observed	No code for screen-capture, blur, or non-road rejection. Photo checks are format/size only (`photoValidation.ts`, magic bytes, ≤ 5 MB). Human review is the only defence.

Field	Content
Acceptance criteria	Detector with documented false-accept/false-reject rates; rejected reports retained for audit, not silently discarded.

F-18 — Overlay and material recommendation are PCI look-ups

Field	Content
Severity / class	P1 · VERIFIED
Affected	`pinnEngine.ts:115-125`
Observed	Thickness $= \max(4, 0.22(75 - \text{PCI}))$ cm when predicted PCI < 70, independent of strain, structure, traffic and material (edge test: 8.1 cm for both 5 cm and 25 cm asphalt). Type string "AC-BC (Binder Course) + AC-WC (Wearing Course) Poly-Modified Asphalt" conflates three products. The blueprint's AC-WC vs PMA cross-simulation (BAB 4.3) is not implemented.
Acceptance criteria	Recommendation from a cited overlay design method, with inputs listed and "advisory, not a construction design" wording.

F-19 — Exported PDF without status disclaimer

Field	Content
Severity / class	P1 · VERIFIED
Affected	`src/services/reporting/pdfGenerator.ts:78-163`
Observed	"Official report" includes OpenVINO latency (l.90), threshold verdicts (l.111), savings and pothole horizon, and the footer "Dihasilkan secara otomatis oleh sistem kecerdasan buatan terpadu PUPR, Kemenhub, & Intel OpenVINO" (l.163), naming institutions that have not endorsed it.
Acceptance criteria	Every page carries engine status, data status and "not an official document"; institution names removed unless authorised.

F-20 — Random attribution percentages

Field	Content
Severity / class	P1 · VERIFIED
Affected	`CommunityReportsDashboard.tsx:259-261` (`estimatedPinnContributionPercent = 72 + Math.random()*20`; summary text "dipercepat hingga X%" uses a second, different random number); `LeafletMap.tsx:360-361`
Acceptance criteria	No contribution percentage without a decomposition method (`docs/AI.md` already states this rule for the new surfaces).

F-21 to F-31 — Medium and low findings

ID	Sev.	Observation and evidence	Acceptance criterion
F-21	P2	Telemetry jitter and `SIM-${Date.now()}` ids make the legacy result payload non-reproducible (edge test). The new assessment path is deterministic (hash id, `assessment.ts:67-74`).	Deterministic payload for identical inputs; test asserts equality.

ID	Sev.	Observation and evidence	Acceptance criterion
F-22	P2	Economic loss $= 0.85 \cdot LF \cdot L$ / 10 (Rp bn), savings $= 0.72 \times$ loss, Zero-ODOL impact $= 1 - 1 / LF$ (negative, e.g. –144 %, below 10 t), "+3.5 years" life extension (`pinnEngine.ts:188-201`); constant budgets in policy views; "Data PUPR 2025" badge on a constant table (`ExecutivePolicyDashboard.tsx:223`). Partially mitigated by the ILUSTRATIF notice on `/policy`.	Cost model with sources, or removal; no official-source badge on constants.
F-23	P2	No status notice on `/engine/simulation`, `/engine/fwd`, `/engine/prediction`, `/engineering`, `/reports`, OperationsHome BERANDA/ALERTS branches, the classic layout and the global modals (subagent sweep; spot-checked).	Lint-style test that every page rendering engine/CV/telemetry output includes a status component.
F-24	P2	Supabase migration (17 tables, RLS) written but not applied; server uses an atomic JSON file store (`fileCaseStore.ts:44-45`). Blueprint `damage_reports` columns (`tensile_strain_et`, `predicted_rul_days`, `pore_water_pressure_kpa`, `is_spoofed`, `substandard_material_flag`) are replaced by `report_cases` + `pinn_predictions` — a defensible normalisation, not recorded against the blueprint. BLOCKED (credentials).	Applied migration, integration tests against it, schema decision in DECISION_LOG.
F-25	P2	Weather is a deterministic hash of the segment id (`evidence.ts:128-142`), labelled DEMO; traffic → load uses an illustrative axle spectrum and 3 axles/vehicle (`evidence.ts:60-69`); no BMKG/OpenWeather/TomTom/HERE integration; no "virtual WIM" regression. Labelling is honest on new surfaces.	Live adapters with provenance, or permanent DEMO labels.
F-26	P2	No "installation year" input or ageing curve (BAB 4.2); no sub-standard material alert (BAB 4.4); no AC-WC/PMA cross-simulation (BAB 4.3).	Implemented with documented logic; alert framed as an investigation trigger, never as proof.
F-27	P2	226 tests pass; physics tests are regression/characterisation only; no independent-reference comparison, no CV model test, no browser E2E (G-SW-02).	Reference-comparison test (from `evidence/compare.mjs`) and a Playwright golden path in CI.
F-28	P2	`assessSegment` clamps inputs (e.g. axle 0 → 10 t, h 0 → 5 cm) and reports strains for the clamped scenario while flagging `outOfRange` (edge test: 73 με for a request of 0 t).	Refuse, or state "computed for clamped values" next to the numbers.
F-29	P3	2D cross-section draws a 15 cm, 300 MPa base the engine ignores and labels total thickness $h + 25$ while drawing $h + 15$ (`SimulationPanel.tsx:468,495`).	Drawing equals model inputs.
F-30	P3	README title "Physics-Informed Neural Network Decision Support System" and "106 tes" (actual 226); UI "Gemini 2.5" (`SimulationPanel.tsx:704`) vs server `gemini-3.7-flash`; package name `react-example`; typo "Mebeg" (`ExecutivePolicyDashboard.tsx:124`).	Documentation matches code; checked in review.
F-31	P3	`Content-Disposition` filename interpolates `req.query.device` without whitelist (`openvinoController.ts`). Node rejects CR/LF in headers, so response splitting is unlikely; quotes can still break the header. NOT VERIFIED by request.	Whitelist of devices and precisions.

A.3 Positive findings (verified strengths)

ID	Strength	Evidence
S-01	Server-enforced workflow state machine with role, assignment, ownership and separation-of-duty rules; audit trail; optimistic concurrency	`src/domain/workflow/*`, `server/routes/workflow.routes.ts`; tests "cross-role golden path", "wrong role, wrong assignee and self-approval are forbidden", "stale versions are rejected", "concurrent transitions are serialized" — all pass

ID	Strength	Evidence
S-02	Honest status vocabulary on the newer surfaces (RULE_BASED, DEMO, SYNTHETIC, VALIDATION_REQUIRED; confidence capped LOW)	`assessment.ts:17,160-195`; `PinnPage.tsx:30-50`; `docs/AI.md`, `docs/validation/PINN_VALIDATION_GAP.md`
S-03	CV → engineering boundary refuses to infer modulus, CBR or load from a photo	`assessment.ts:238-257`
S-04	DEMO CV can never auto-verify a report	`screening.ts:88-97`; test passes
S-05	ESAL arithmetic correct and input-validated	`esal.ts:30-51`
S-06	What-If runs persisted with physical-direction sanity checks and provenance	`whatIf.ts`, `intelligence.routes.ts:95-125`; test "What-If runs are persisted as SIMULATED records"
S-07	Atomic persistence (temp file + rename) and upload magic-byte checks	`fileCaseStore.ts:44-45`; `photoValidation.ts`
S-08	Prior internal validation-gap report already identified the engine's deterministic identity and similarity failures; this audit independently reproduces and extends it with a reference solver	`docs/validation/PINN_VALIDATION_GAP.md`

Appendix B — Mathematical Verification Register (Artifact C)

Document PAVEMENT-PINN-AUDIT-001 · Version 1.0 · Commit `4a9b4fc`. Equation provenance codes: **BP** = specified in the master blueprint, **SRC** = found in source code, **AUD** = independently derived or implemented during this audit, **NUM** = numerically verified, **UNV** = validity unverified.

B.1 Equation inventory

Table B.1 — Equation inventory and verification status

Eq.	Relationship	Provenance	Location	Units	Verification method	Result
EQ-1	PINN total loss (four weighted terms)	BP	Blueprint BAB 5.1	dimensionless	Search for loss, optimiser, autodiff	NOT IMPLEMENTED
EQ-2	Equilibrium $\nabla \cdot \sigma + \mathbf{b} = 0$	BP; AUD	Blueprint BAB 5.1; satisfied by construction in `evidence/mlet.mjs`	MPa/m	Code search (engine); reference solver checks	Engine: NOT IMPLEMENTED; reference: VERIFIED
EQ-3	Constitutive $\sigma = \mathbf{C}{:}\,\varepsilon$ (isotropic)	BP; AUD	Blueprint BAB 5.1; `mlet.mjs` strain recovery	MPa	Reference kernel consistency (B.4)	Engine: NOT IMPLEMENTED; reference: VERIFIED
EQ-4	Engine tensile strain ε_t	SRC	`pinnEngine.ts:58-62`	με (integer); P in t, h in cm, E in MPa	Re-derivation; existing test; reference comparison	Arithmetic VERIFIED; physics FAILED
EQ-5	Engine compressive strain ε_v	SRC	`pinnEngine.ts:65-69`	με	as EQ-4	Arithmetic VERIFIED; physics FAILED
EQ-6	Engine surface deflection w_0 and basin	SRC	`pinnEngine.ts:72-88`	μm	Similarity test; basin shape	Arithmetic VERIFIED; physics FAILED
EQ-7	Moisture decay and $E_s = 10$ CBR	SRC	`pinnEngine.ts:37-43`	–, %, MPa	Source search	UNVERIFIED assumption
EQ-8	Asphalt temperature softening	SRC	`pinnEngine.ts:46-50`	°C, MPa	Source search	UNVERIFIED assumption
EQ-9	Load factor $(P/10)^4$ vs LEF $(P/8.16)^4$	SRC	`pinnEngine.ts:54`; `esal.ts:12,30-35`	–	Arithmetic; cross-module comparison	Arithmetic VERIFIED; consistency FAILED
EQ-10	PCI drop, predicted PCI, remaining life, pothole horizon	SRC	`pinnEngine.ts:92-112`	index, years, days	Edge cases; structural sensitivity	FAILED (decoupled, non-finite at P=0)
EQ-11	Overlay thickness	SRC	`pinnEngine.ts:115-118`	cm	Structural sensitivity	UNVERIFIED heuristic
EQ-12	Love strain function relations (axisymmetric)	AUD	`evidence/mlet.mjs`	–	Boussinesq limit, kernel consistency	VERIFIED (NUM)
EQ-13	Boussinesq on-axis closed form	AUD (textbook)	`evidence/mlet.mjs boussinesq()`	MPa, m	Agreement with solver ≤ 1e-12 relative	VERIFIED (NUM)

Eq.	Relationship	Provenance	Location	Units	Verification method	Result
EQ-14	Elastic similarity $\varepsilon(kE) = \varepsilon(E) / k$	AUD	B.4	–	Reference: exact; engine: exponents	Reference VERIFIED; engine FAILED
EQ-15	Engine similarity exponents	AUD	derived from EQ-4..6	–	Analytical	VERIFIED (−0.4, −0.75, −0.6)
EQ-16	Transfer-function form $N_f = k_1 \varepsilon_t^{-k_2} E^{-k_3}$	AUD (form only)	not in code	repetitions	Search for traffic-dependent criterion	NOT IMPLEMENTED

B.2 Equations specified in the blueprint (BLUEPRINT REFERENCE)

$$\mathcal{L}_{\text{total}} = \lambda_d \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{physics}} + \lambda_{bc} \mathcal{L}_{bc} + \lambda_{ic} \mathcal{L}_{ic} \quad \text{(EQ-1)}$$

Blueprint BAB 5.1 total loss. Not present in source code.

$$\nabla \cdot \sigma + \mathbf{b} = 0 \quad \text{(EQ-2)}$$

Mechanical equilibrium (quasi-static). Not evaluated by the engine.

$$\sigma = \mathrm{C} : \varepsilon, \quad \varepsilon = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) \quad \text{(EQ-3)}$$

Linear elastic constitutive law. Not evaluated by the engine.

The blueprint also names an initial-condition loss for "transient rain infiltration". The engine is quasi-static and time-independent, so no initial condition exists to enforce; a transient moisture model would need its own governing equation (e.g. Richards' equation or a consolidation model), which the blueprint does not specify.

B.3 Equations found in source code (AS-IMPLEMENTED)

Inputs: axle load P (t), asphalt thickness h (cm), asphalt modulus E_1 (MPa), subgrade CBR (%), rain intensity R (mm/h), flood duration t_f (h), surface temperature T (°C), current PCI of the segment.

$$f_w = \max\left(0.4, 1 - 0.004R - 0.015 t_f\right), \quad E_s = 10 \cdot \mathrm{CBR} \cdot f_w \quad \text{(EQ-7)}$$

Moisture decay of CBR and subgrade modulus (pinnEngine.ts:37-43).

$$E_1' = E_1 \cdot \max\left(0.5, 1 - 0.012\left(T - 25\right)\right) \quad \text{(EQ-8)}$$

Temperature softening of asphalt modulus (pinnEngine.ts:46-50).

$$\varepsilon_t = \mathrm{round}\left(\frac{120P}{h^{1.75}\left(E_1' / 2000\right)^{0.4}}\right) \quad \text{(EQ-4)}$$

Tensile strain at the bottom of the asphalt, microstrain, rounded (pinnEngine.ts:58-62).

$$\varepsilon_v = \mathrm{round}\left(\frac{210P}{\left(h + 15\right)^{1.4}\left(E_s / 40\right)^{0.75}}\right) \quad \text{(EQ-5)}$$

Vertical compressive strain at the top of the subgrade, microstrain (pinnEngine.ts:65-69).

$$w_0 = \text{round}\left(\frac{45P}{\left(E_s / 40 \right)^{0.6} \left(h / 10 \right)^{0.5}} \right), \quad w\left(r \right) = w_0 e^{-0.0018r} \tag{EQ-6}$$

Surface deflection (μm) and fixed exponential basin, r in mm (pinnEngine.ts:72-88).

$$LF = \left(\frac{P}{10} \right)^4, \quad LEF = \left(\frac{P}{8.16} \right)^4$$
$$\frac{LEF}{LF} = \left(\frac{10}{8.16} \right)^4 = 2.255 \tag{EQ-9}$$

Engine load factor vs ESAL load equivalency factor (pinnEngine.ts:54; esal.ts:30-35).

$$\Delta_{PCI} = \min\left(35, \frac{2.5LF}{f_w} \right), \quad PCI' = \max\left(10, PCI_0 - 0.8\Delta_{PCI} \right)$$
$$RUL = \max\left(0.3, \frac{PCI'}{1.2\Delta_{PCI}} \right), \quad H_{pothole} = \max\left(3, \frac{1800 f_w}{LF} \right) \tag{EQ-10}$$

Deterioration rules (pinnEngine.ts:92-112).

$$h_{ov} = \max\left(4, 0.22\left(75 - PCI' \right) \right) \tag{EQ-11}$$

Overlay recommendation (pinnEngine.ts:115-118), applied when PCI' < 70.

No equation in EQ-10 or EQ-11 contains ε_t, ε_v, h, E_1 or CBR: the structural response does not influence life, horizon or overlay (verified numerically, B.6).

B.4 Independent reference solution (AUD, NUM)

Formulation. Axisymmetric multilayer linear elasticity with fully bonded interfaces, uniform circular pressure q of radius a, half-space bottom layer, solved in the Hankel domain with Love's strain function $\varphi\left(r, z \right) = J_0\left(mr \right) f\left(z \right)$, $f = \left(A + Bz \right) e^{mz} + \left(C + Dz \right) e^{-mz}$ in local per-layer coordinates. The relations used (Timoshenko & Goodier, Love strain function) are:

$$2Gu_z = 2\left(1 - \nu \right)\nabla^2\varphi - \frac{\partial^2\varphi}{\partial z^2}$$
$$\sigma_z = \frac{\partial}{\partial z}\left[\left(2 - \nu \right)\nabla^2\varphi - \frac{\partial^2\varphi}{\partial z^2} \right]$$
$$\tau_{rz} = \frac{\partial}{\partial r}\left[\left(1 - \nu \right)\nabla^2\varphi - \frac{\partial^2\varphi}{\partial z^2} \right] \tag{EQ-12}$$

Love strain-function relations used by the reference solver (tension positive, z downward).

Boundary conditions: $\sigma_z\left(r, 0 \right) = -q$ for $r \le a$ (Hankel amplitude $-qaJ_1\left(ma \right)$), $\tau_{rz}\left(r, 0 \right) = 0$; continuity of $u_z, u_r, \sigma_z, \tau_{rz}$ at each interface; decay at depth. Strains are recovered from stresses by Hooke's law (EQ-3) and, independently, from the displacement kernel; both must agree.

$$\sigma_z = -q\left[1 - \frac{z^3}{\left(a^2 + z^2 \right)^{3/2}} \right], \quad w_0 = \frac{2\left(1 - \nu^2 \right) qa}{E} \tag{EQ-13}$$

Boussinesq on-axis closed form (flexible circular load) used as the analytical limit.

$$\varepsilon\left(kE_1, kE_2, \ \ldots\right) = \frac{1}{k}\varepsilon\left(E_1, E_2, \ \ldots\right) \quad \text{(EQ-14)}$$

Elastic similarity: a necessary property of every linear-elastic layered solution.

$$\varepsilon_t \propto k^{-0.4}, \quad \varepsilon_v \propto k^{-0.75}, \quad w_0 \propto k^{-0.6} \quad \text{(EQ-15)}$$

Engine exponents under the same scaling (derived from EQ-4 to EQ-6).

Table B.2 — Verification of the reference solver

Check	Expected	Result	Status
Bessel J_0, J_1 at tabulated points (x = 1, 10, first zero of J_0)	abs. error ≤ 1e-8	met	VERIFIED
Homogeneous 3-layer system vs Boussinesq: $\sigma_z, \sigma_r, \varepsilon_z, \varepsilon_r, w$ at 6 depths on both sides of each interface	rel. error ≤ 2e-3	max ≈ 2e-12	VERIFIED
Surface deflection vs $2\left(1-\nu^2\right)qa\,/\,E$	rel. error ≤ 5e-3	1.0e-5	VERIFIED
Similarity: all moduli × 3	strain ratio 3	3.0000000	VERIFIED
Layered system (3000/250/50 MPa): strain from displacement kernel = strain from stress kernel	rel. ≤ 1e-6	≤ 2e-14	VERIFIED
Continuity of $\sigma_z, w, \varepsilon_r$ across interfaces	equal	equal to 1e-9	VERIFIED
Burmister two-layer interface stress $\sigma_z\,/\,q, h_1\,/\,a = 1, \nu = 0.5$: chart values ≈ 0.646 / 0.29 / 0.08 for $E_1\,/\,E_2$ = 1 / 10 / 100	chart reading	0.646 / 0.291 / 0.081	VERIFIED (chart-reading accuracy)
Comparison with a certified program (KENLAYER, WESLEA, PyMastic, BISAR)	—	not available on the audit machine	BLOCKED

Table B.3 — Engine vs reference over the in-range grid (144 points; P 10-25 t, h 5-25 cm, E_1 1200-3500 MPa, CBR 3-10; dry, 25 °C; dual-tyre single axle as four isolated tyres at 0.55 MPa; ν = 0.35 asphalt and base, 0.45 subgrade; $E_s = 10$ CBR as in the engine)

Ratio engine / reference	min	p25	median	p75	max
εt, two-layer (asphalt on subgrade)	0.020	0.036	0.047	0.073	0.323
εt, three-layer (+15 cm base, 200 MPa)	0.029	0.047	0.067	0.137	0.786
εv, two-layer	0.007	0.013	0.019	0.026	0.062
εv, three-layer	0.013	0.020	0.025	0.033	0.063

Table B.4 — Blueprint default segment (BAB 8.2: AC-WC 10 cm, E 2500 MPa, CBR 6), dry, 25 °C

Axle	Loading assumption	Engine εt / εv (με)	Reference 2-layer εt / εv	Reference 3-layer εt / εv
8.16 t	4 tyres, 0.55 MPa (a = 10.8 cm)	16 / 14	387 / 1029	274 / 663
10 t	4 tyres, 0.55 MPa (a = 11.9 cm)	20 / 17	435 / 1174	303 / 793
10 t	2 tyres, 0.70 MPa (a = 14.9 cm)	20 / 17	703 / 1968	472 / 1475
22.5 t	4 tyres, 0.55 MPa (a = 17.9 cm)	44 / 38	652 / 1889	424 / 1536
22.5 t	4 tyres, 0.70 MPa (a = 15.8 cm)	44 / 38	744 / 2106	494 / 1622

Table B.5 — Local log-sensitivities $\partial \ln \varepsilon / \partial \ln x$ at the blueprint point (reference two-layer vs engine formula)

Variable	εt reference	εt engine	εv reference	εv engine
Axle load P (constant contact pressure)	0.56	1.00	0.64	1.00
Asphalt thickness h	−1.12	−1.75	−1.27	−0.56
Asphalt modulus E_1	−0.68	−0.40	−0.63	0.00
Subgrade CBR	−0.32	0.00	−0.37	−0.75

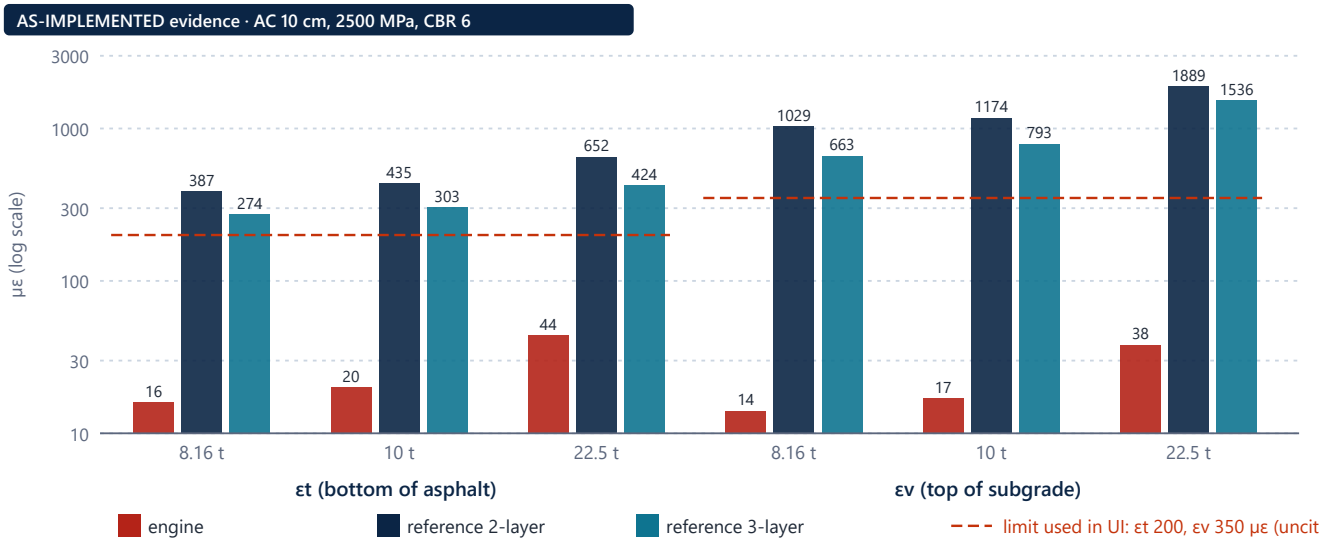


Figure B.1 — Engine vs reference strains for the blueprint default segment (log scale). AS-IMPLEMENTED evidence; reference assumptions as in Table B.4.

Interpretation. The engine/reference gap is far larger than any plausible variation of the reference assumptions. Halving the reference wheel load changes its strains by a factor of about $2^{0.6} \approx 1.5$; the gap is typically a factor of about 14 to 77 (the inter-quartile ranges of Table B.3, inverted) and up to about 140; it narrows only for εt in thin (5 cm), low-modulus (1200 MPa) asphalt on the three-layer structure, where the 200 MPa base limits the reference tensile strain (maximum ratio 0.79 at 25 t; εv ratio at the same point 0.02). Dual-wheel interaction would increase the reference εv further (superposition), widening the gap. The conclusion "engine strains are one to two orders of magnitude below linear-elastic layered theory" is therefore robust to the loading and Poisson-ratio assumptions. It does **not** establish what the true field strains are; that requires measured or certified-model data (B.8).

B.5 Strain threshold and failure-criterion register

Table B.6 — Threshold register (no value changed during the audit)

Parameter	Source location	Value	Physical meaning claimed	Source/standard cited	Verification status	Risk
εt "WASPADA"	`pinnEngine.ts:130`	180 με	approaching fatigue threshold	none	UNVERIFIED	Medium

Parameter	Source location	Value	Physical meaning claimed	Source/standard cited	Verification status	Risk
εt "permitted limit"	`pinnEngine.ts:129` (text), `FieldEngineerTools.tsx:198`, `FieldEngineerDashboard.tsx`, `SimulationPanel.tsx:563-571`, `pdfGenerator.ts:111`	200 με	fatigue limit, labelled "AASHTO" (variable, legend) and "Bina Marga" (legend, engine text)	none (two different bodies named)	UNVERIFIED	High
εt "KRITIS" trigger	`pinnEngine.ts:128`	250 με	exceeds permitted limit (text says 200)	none	FAILED internal consistency	High
εv "permitted limit"	`FieldEngineerTools.tsx:203`, `FieldEngineerDashboard.tsx`	350 με	rutting limit	none	UNVERIFIED; unreachable by engine (max 98 με on grid)	High
Placeholder εt / εv	`FieldEngineerTools.tsx:207-208`	245 / 380 με	shown when no result	n/a	FAILED (fabricated display values)	Medium
"+22.5 %" exceedance	`FieldEngineerTools.tsx:397`	constant	fatigue status	n/a	FAILED (hard-coded)	High
Alert RED/YELLOW	`pinnEngine.ts:136-140`	P > 18 / 14 t, R > 50 / 25 mm/h, PCI' < 40 / 60	operational alert	none	UNVERIFIED	Medium
Dynamic tonnage limit	`pinnEngine.ts:143-148`	18 − 0.08 R − 0.2 t_f tonnes, floor 10 t	weather-adjusted legal load	none	UNVERIFIED	High (enforcement wording)
Screening confidence	`stateMachine.ts` (`SCREENING_CONFIDENCE_THRESHOLD`)	constant	CV auto-pass	design choice	n/a until a REAL model exists	Low

Distinctions the code does not make: (1) elastic strain under one load versus accumulated permanent deformation; (2) fatigue life versus rutting life — no repetitions-to-failure is computed; (3) structural failure versus visible pothole formation — "pothole horizon" is a load/rain rule (EQ-10), not a failure criterion. Required to close: the governing document and clause for each limit (the team should state whether Bina Marga MDP 2017 or AASHTO is adopted), the design traffic (ESAL) to which a fixed limit corresponds, or replacement by transfer functions of the EQ-16 form calibrated or adopted from that document.

$$N_f = k_1 \varepsilon_t^{-k_2} E_1^{-k_3}, \quad N_d = k_4 \varepsilon_v^{-k_5} \tag{EQ-16}$$

General form of mechanistic-empirical transfer functions (form only; coefficients must come from the adopted standard).

B.6 Boundary and edge-case results (engine function, dry 25 °C baseline: 10 t, 10 cm, 2500 MPa, CBR 6, PCI 80)

Table B.7 — Edge cases (`evidence/edge.mjs`; non-finite values appear as `null` after JSON serialisation)

Case	εt / εv / w_0	PCI' / RUL (yr) / horizon (d)	Other	Classification
Baseline 10 t	20 / 17 / 353	78 / 26 / 1800	GREEN	reference

Case	εt / εv / w_0	PCI' / RUL (yr) / horizon (d)	Other	Classification
Zero load	0 / 0 / 0	80 / ∞ / ∞	Zero-ODOL impact −∞	FAILED non-finite
Negative load (−5 t)	−10 / −9 / −176	80 / 333.3 / 28 800	Zero-ODOL −1500 %	FAILED accepted
8 t (below reference)	16 / 14 / 282	79 / 65.8 / 4395	Zero-ODOL −144 %	FAILED negative percentage
100 t	195 / 171 / 3528	52 / 1.2 / 3	RED	outside documented range; accepted
Thickness 0	∞ / 35 / ∞	unchanged	—	FAILED
CBR 0 or negative	20 / ∞ / ∞	unchanged	—	FAILED
Modulus 0	∞ / 17 / 353	unchanged	—	FAILED
Rain 1000 mm/h	20 / 34 / 611	75 / 9.9 / 720	RED	floor 0.4 masks input error
Temperature −40 °C / 200 °C	15 / 17 / 353 · 26 / 17 / 353	unchanged	—	linear softening unbounded below 25 °C
`NaN` load	NaN throughout	NaN	GREEN alert	FAILED silent
Missing thickness	NaN	unchanged	—	FAILED silent
String "12"	coerced to 12	—	—	FAILED type coercion
18 t with h 5 vs 25 cm, CBR 3 vs 10, E 1200 vs 3500, T 60 °C	—	59 / 1.9 / 171 in every case	overlay 8.1 cm for h 5 and 25 (PCI 40 segment)	FAILED structure has no effect
Same input twice	identical strains	identical	latency 34.1 vs 34.2 ms; physics loss 0.00185 vs 0.00227	FAILED non-deterministic telemetry

B.7 Dimensional and unit traceability

Table B.8 — Units through input, calculation, storage, API and display

Quantity	Input	Calculation	Storage (migration)	API	Display	Finding
Axle load	t (slider 10-25)	used as t in fitted power laws	`input_snapshot` jsonb	t	"Ton"	Force unit implicit; not converted to kN; no wheel load
Contact stress	—	$0.65P$ / 10 MPa	jsonb	MPa	MPa	Proportional to load (non-physical for tyre contact)
Elastic modulus	MPa	ratio E / 2000	`modulus_min/max_mpa`	MPa	MPa	consistent unit; exponent non-physical
Thickness	cm	$h^{1.75}$, $(h+15)^{1.4}$ (cm)	`surface_thickness_cm`	cm	cm	hidden 15 cm term (base?) undocumented
Strain	—	με, rounded to integer	`outputs` jsonb	με	με	rounding loses resolution below 1 με (immaterial at present scale)
Deflection	—	μm	jsonb	μm	μm / basin mm	basin shape fixed
Rainfall	mm/h	linear decay factor	weather observation	mm/h	mm/jam	intensity used as moisture state
CBR	%	$E_s = 10$ CBR MPa	`subgrade_cbr_min/max`	%	%	correlation source not cited; validity typically limited to low CBR

Quantity	Input	Calculation	Storage (migration)	API	Display	Finding
Pore-water pressure	—	absent	absent	absent	particle animation in blueprint only	NOT IMPLEMENTED
Time	—	years, days	jsonb	years /days	"Hari", "Thn"	three inconsistent time-to-failure definitions
PCI	index 0-100	heuristic projection	`outputs` jsonb	index	index	PCI definition (distress survey) not respected
Latency	—	constant + random	—	ms	ms	not a measurement

B.8 Limitations of the mathematical verification

1. The reference solver is an audit implementation verified against closed forms and chart readings, not a certified program. A cross-check against KENLAYER, WESLEA, BISAR or PyMastic is recommended before it is used as a design tool (R-P3-01).
2. Loading assumptions (tyre count, 0.55-0.70 MPa, isolated tyres) and Poisson ratios are stated assumptions; the conclusion is robust to them (B.4) but the reference values are not field predictions.
3. $E_s = 10$ CBR was retained only to compare like with like; it is itself an unverified correlation (EQ-7).
4. No empirical validation (FWD basins, strain gauges, longitudinal PCI) was possible: no such data exists in the repository. Empirical accuracy of both engine and reference remains NOT VERIFIED.

Appendix C — Test and Benchmark Evidence (Artifact E)

Document PAVEMENT-PINN-AUDIT-001 · Version 1.0. Environment: Windows 11 Home 10.0.26200, Node.js v24.16.0, npm scripts from `package.json`, Chrome 155.0.8059.40 (PDF rendering only). Python is not installed on the audit machine (`python` resolves to the Microsoft Store stub), so no PyTorch, Ultralytics or OpenVINO command could run. All commands were run from the repository root unless stated otherwise. No command modified tracked files, the database, `dist/` or the user's development server.

C.1 Test execution register

Table C.1 — Commands executed

Test ID	Command or procedure	Environment	Result	Evidence	Limitation
T-01	`git rev-parse`, `git log --oneline -15`, `git status --short`, `git stash list`, `git worktree list`	repo root	HEAD `4a9b4fc`, branch `feature/m1-report-ownership`; 24 modified, 1 deleted, 10 untracked paths; no stash; one worktree	Document Control	Read-only snapshot at audit start
T-02	`netstat -ano` filtered for ports 3000/3100/3200	host	Port 3000 LISTENING (PID 18032, the user's server); 3100/3200 free	Document Control	Server was not called, restarted or reconfigured
T-03	`npm test` (`tsx --test "tests/**/*.test.ts"`)	working tree incl. uncommitted changes	226 tests, 226 pass, 0 fail, exit 0, 14.3 s	console summary	17 of the 226 tests live in untracked files of another session (`auth-api`, `demo-auth`); the committed HEAD alone was not tested separately
T-04	`npm run lint` (`tsc --noEmit`)	working tree	exit 0, no diagnostics	console	Type safety only
T-05	`node evidence/verify-mlet.mjs`	audit folder	PASS: Bessel values; homogeneous 3-layer = Boussinesq (max rel. error ≈ 2e-12); surface deflection rel. error 1.0e-5; similarity ratio 3.0000000	`evidence/evidence-run.log`	Verifies the audit's reference solver only
T-06	`node evidence/verify-layered.mjs`	audit folder	PASS: kernel consistency ≤ 2e-14; interface continuity; Burmister chart values 0.646/0.291/0.081	`evidence/evidence-run.log`	Chart-reading accuracy for the Burmister comparison
T-07	`tsx evidence/compare.mjs`	audit folder, imports engine read-only	exit 0; Tables B.3-B.5	`evidence/compare-grid.json`, log	Loading assumptions stated in B.4
T-08	`tsx evidence/edge.mjs`	audit folder, imports engine read-only	exit 0; Table B.7	`evidence/evidence-run.log`	Function-level; HTTP path not exercised
T-09	Static UI-claim sweep (subagent, read-only grep) + manual spot-check of 10 cited locations	`src/`	All 10 spot-checks matched	Appendix A references	Automated sweep may miss dynamically built strings
T-10	Blueprint text extraction `pdftotext -layout`	scratch copy of the blueprint PDF	1062 lines, BAB 1-10	—	Some mathematical glyphs are lost in the text layer; blueprint page images could not be rendered (no `pdftoppm`)

C.2 Tests deliberately not performed

Table C.2 — Not performed, with reason

Item	Reason	Status
`npm run build`	Writes `dist/`, which already holds a build from 2026-10-14 16:31 that the user may rely on; not needed to establish any finding	Not run (safety)
Second dev server or production smoke test	Not required for findings; would add processes beside the user's server	Not run
HTTP calls to `/api/simulate`, `/api/openvino/*`	Port 3000 belongs to the user; function-level evidence suffices	Not run
Browser end-to-end golden path	Prior session reports a manual run (`docs/FINAL_AUDIT.md`); no Playwright suite exists (G-SW-02)	NOT VERIFIED by this audit
Camera / GPS on a phone	No device available	BLOCKED
CV model evaluation (mAP, per-class)	No weights, no local dataset, no Python	BLOCKED
PINN training, ablation, held-out evaluation	No model, no data, no Python	BLOCKED
OpenVINO conversion, quantisation, latency	No model, no runtime, no Python	BLOCKED
Comparison with a certified layered-elastic program	None installed	BLOCKED
FWD / field validation	No data in repository	BLOCKED

C.3 Classification of the existing test suite

Table C.3 — What each existing suite establishes

Category	Files (tests)	Establishes	Does not establish
1. Unit	`tests/unit/*` (144)	Workflow rules, RBAC, report factory and ownership, maps, routes, module boundaries, type gate, intelligence helpers	Physical validity
2. Integration	`tests/integration/*` (49)	HTTP contracts, auth tokens, hardening, workflow and intelligence endpoints	Physical validity; real IdP or database
3. Mathematical verification	`pinn-mathematical-audit.test.ts` (14)	Code equals the documented closed-form formulas; monotonicity; known deviations pinned	Agreement with elasticity or measurement
4. Numerical benchmark	`pinn-regression.test.ts` (14), `pinn-threshold-characterization.test.ts` (5)	Output unchanged against 76 golden scenarios; threshold reachability	Correctness of the golden values
5. Empirical validation	none	—	—
6. Production / hardware benchmark	none	—	—

Counts from `grep -c "test("` per file: `tests/unit` 144, `tests/integration` 49, `tests/physics-validation` 33; total 226, matching the runner. Of these, 17 tests (`auth-api` 8, `demo-auth` 9) are in files untracked by Git (another session's work in progress).

C.4 Edge benchmark register

Table C.4 — Edge benchmark register (no measurement exists)

Benchmark	Device	Precision	Latency	Numerical deviation	Status
PINN inference per scenario (target < 10 ms)	CPU / Arc iGPU / NPU	FP32 / FP16 / INT8	not measured	not measured	BLOCKED (no model, no runtime)
YOLOv8 inference	CPU / iGPU	INT8	not measured	not measured	BLOCKED
SLM (Phi-3-mini / Llama-3.2)	NPU	INT4	not measured	not measured	NOT IMPLEMENTED
UI "benchmark" (`simulatedBenchmark.ts`)	none (lookup table + JS loop)	n/a	constants 6.8-22.5 ms × factor	constant 99.4-100 %	SIMULATED — not a measurement
Engine formula evaluation in JavaScript	host CPU (Node)	FP64	not timed (sub-millisecond by inspection: ~40 arithmetic operations)	n/a	NOT VERIFIED; irrelevant to OpenVINO

Required benchmark protocol before any latency claim (Appendix F of the report): fixed model hash; OpenVINO version; `core.available_devices` output; device driver versions; precision and NNCF configuration; batch size and input shape; 50 warm-up runs; ≥ 1000 timed runs; cold-start time; p50/p95/p99; throughput; peak resident memory; maximum absolute and relative deviation of outputs vs FP32 on a fixed validation set; power measured with a stated tool, or reported as not measured.

Appendix D — Blueprint Traceability Matrix (Artifact D)

Reference: *Master Blueprint & Kanvas Arsitektur Lengkap PAVEMENT-PINN (Versi Final Terbaru)*, 25 pages, chapters BAB 1–10 (read from the PDF text layer). Implementation: commit `4a9b4fc` plus working tree, 2026-10-14. Status vocabulary: VERIFIED implemented and demonstrated; PARTIALLY VERIFIED partly implemented or demonstrated; NOT IMPLEMENTED; NOT VERIFIED; BLOCKED; DIVERGENT implemented differently from the blueprint by a recorded or apparent design decision; N/A narrative content with no software obligation. A planned capability, a UI depiction or a formula that produces a number is not counted as implemented.

Table D.1 — Requirement-by-requirement traceability

Req. ID	Blueprint ref.	Expected behaviour	Implementation location	Evidence	Status	Gap
BP-1.1	BAB 1.1	National road-condition, budget and ODOL statistics	— (narrative)	Figures not re-verified by the audit; some reappear as constants in policy views	N/A	Cite sources wherever figures are shown in the UI (F-22)
BP-1.2a	BAB 1.2 Fase 1	Pilot corridor Jalan Raya Narogong, Bekasi–Bogor	`src/data/reference/karawangPilot.ts` (Karawang KRW-001..004); `screening.ts:40`	No "Narogong" in `src/`; Narogong-area report → OUT_OF_SCOPE (edge test)	DIVERGENT	Governing document undecided (F-15)
BP-1.2b	BAB 1.2 Fase 2	Regional expansion (Klari–Simpang Jomin, Kaligawe) and toll WIM	Karawang data used as the pilot	Code comment cites spec v2.2 §6	DIVERGENT	Phase order inverted relative to blueprint
BP-1.3	BAB 1.3	SDG 9, 11, 12, 13, 17 alignment	—	Narrative	N/A	—
BP-2.1	BAB 2.1	FWD data as Data Loss; FEM-equivalent solutions in < 10 ms	—	No FWD data, no FEM, no model	NOT IMPLEMENTED	F-01, F-14, F-16
BP-2.2-N1	Novelty 1	Predictive-preventive paradigm via continuous physics simulation	`pinnEngine.ts`; `assessment.ts`	Formula engine, not continuous simulation	PARTIALLY VERIFIED	F-03, F-07
BP-2.2-N2	Novelty 2	Data Loss + Physics Loss PINN	—	No training code	NOT IMPLEMENTED	F-01
BP-2.2-N3	Novelty 3	εt, εv and pore-water pressure computed instantly	`pinnEngine.ts:58-69`	εt, εv by fitted formulas (1-2 orders below elastic reference); no pore pressure	PARTIALLY VERIFIED	F-03, F-10
BP-2.2-N4	Novelty 4	Virtual WIM + anti-spoofing CV	`evidence.ts` (illustrative spectrum)	No traffic API, no regression, no anti-spoofing	NOT IMPLEMENTED	F-17, F-25
BP-2.2-N5	Novelty 5	Material recommender + quality audit	`pinnEngine.ts:115-125`	PCI look-up; no audit logic	NOT IMPLEMENTED	F-18, F-26
BP-2.2-N6	Novelty 6	One inference → three role outputs	`pinnEngine.ts` `roleOutputs`	One function returns three role blocks	VERIFIED (software)	Outputs inherit F-03/F-07
BP-2.2-N7	Novelty 7	RBAC as information translation layer	`domain/workflow`, `features/rbac`, `routeAccess.ts`	RBAC tests pass (226/226 suite)	VERIFIED (demo identity)	Real IdP BLOCKED
BP-2.2-N8	Novelty 8	Heterogeneous CPU + Arc iGPU + NPU execution	—	Simulated telemetry only	NOT IMPLEMENTED	F-02, F-14
BP-2.2-N9	Novelty 9	NNCF INT8/FP16 quantisation preserving PDE precision	—	No model, no NNCF	NOT IMPLEMENTED	F-14
BP-2.2-N10	Novelty 10	On-device SLM (Phi-3/Llama-3.2) on NPU, no third-party data flow	`narrativeService.ts:9` (cloud Gemini) + template fallback	Cloud API, not on-device	DIVERGENT	F-15

Req. ID	Blueprint ref.	Expected behaviour	Implementation location	Evidence	Status	Gap
BP-3.F1a	BAB 3 Fase 1	YOLOv8 trained on GRDDC + Indonesian data	— (`docs/AI.md` audits an external Kaggle notebook, single class, no weights)	No weights in repository	NOT IMPLEMENTED	F-05, F-16
BP-3.F1b	BAB 3 Fase 1	PINN trained with FWD/FEM data loss + Burmister physics loss	—	—	NOT IMPLEMENTED	F-01
BP-3.F1c	BAB 3 Fase 1	PyTorch → ONNX → OpenVINO IR → INT8/FP16 (NNCF)	`openvinoExportService.ts` (hand-written XML)	Illustrative artifacts only	NOT IMPLEMENTED	F-13, F-14
BP-3.F1d	BAB 3 Fase 1	Auto-fill baseline (SNI Bina Marga) per segment in Supabase	`evidence.ts:165-181` `pavementBaseline`; `pavement_baselines` table	Synthetic baseline in code; DB not connected	PARTIALLY VERIFIED	F-24; SNI source not cited
BP-3.F2a	BAB 3 Fase 2	Zero-install PWA, WebRTC camera, Geolocation < 15 s, no registration	`ReportRoadIssueCard.tsx`, `ModernAppLayout.tsx` (getUserMedia/geolocation); anonymous session	Anonymous reporting tested; camera/GPS not exercised on a device; no PWA manifest/QR verified	PARTIALLY VERIFIED	Device test; PWA manifest
BP-3.F2b	BAB 3 Fase 2	Automatic weather pull (BMKG/OpenWeather) by report GPS	`evidence.ts:128-153`	Deterministic DEMO provider	NOT IMPLEMENTED	F-25
BP-3.F2c	BAB 3 Fase 2	Virtual WIM from traffic API speeds + regression to axle load/ESAL	`evidence.ts:60-110`, `esal.ts`	ESAL arithmetic correct; inputs illustrative; no speed-based inference	PARTIALLY VERIFIED	F-25; speed ≠ weight (blueprint assumption unvalidated)
BP-3.F3a	BAB 3 Fase 3	Anti-spoofing auto-reject	—	Format/size checks only	NOT IMPLEMENTED	F-17
BP-3.F3b	BAB 3 Fase 3	Damage class + severity + confidence (Bina Marga classes)	`cvDamageClasses.ts`	Keyword match, constant confidence	NOT IMPLEMENTED	F-05
BP-3.F3c	BAB 3 Fase 3	Visual label → E_1 penalty (e.g. 40-60 %)	`assessment.ts:238-257` refuses mapping	Deliberately not implemented (calibration required)	NOT IMPLEMENTED	Correct pending calibration data
BP-3.F3d	BAB 3 Fase 3	Rain → pore pressure, E_1 penalty 20-30 %, CBR 6 → 2.5 %	`pinnEngine.ts:37-43`	Linear CBR decay, no pore pressure, no bitumen penalty	PARTIALLY VERIFIED	F-10
BP-3.F4a	BAB 3 Fase 4	εt (bottom AC), εv (top subgrade) from PINN on OpenVINO	`pinnEngine.ts:58-69`	Formula outputs; no OpenVINO	PARTIALLY VERIFIED	F-03, F-14
BP-3.F4b	BAB 3 Fase 4	RUL / pothole horizon in days	`pinnEngine.ts:103-112`	Heuristic; non-finite at zero load	PARTIALLY VERIFIED	F-07
BP-3.F4c	BAB 3 Fase 4	PCI projection 0-100	`pinnEngine.ts:92-100,173-186`	Heuristic	PARTIALLY VERIFIED	F-07

Req. ID	Blueprint ref.	Expected behaviour	Implementation location	Evidence	Status	Gap
BP-3.F4d	BAB 3 Fase 4	Inference < 10 ms	`simulatedBenchmark.ts`	Not measured	NOT VERIFIED	F-14
BP-3.F5	BAB 3 Fase 5	Rule engine + local SLM → role recommendations	`pinnEngine.ts` rules; `explain.ts`; `narrativeService.ts`	Rule engine present; SLM absent	PARTIALLY VERIFIED	F-15
BP-4.1	BAB 4.1	Material catalogue (AC-WC, AC-BC, AC-Base, PMA, SMA, HRS) with E ranges	`TechnicalReferenceView.tsx` (static text)	Not used by any computation	PARTIALLY VERIFIED	Engine has one asphalt layer only
BP-4.2	BAB 4.2	Ageing curve from last installation year	—	No installation-year input	NOT IMPLEMENTED	F-26
BP-4.3	BAB 4.3	AC-WC vs PMA cross-simulation; overlay ≥ 8.5 cm directive	`pinnEngine.ts:115-125`	Not implemented; "8.5 cm" appears as a UI constant	NOT IMPLEMENTED	F-18
BP-4.4	BAB 4.4	Sub-standard material anomaly alert (no "korupsi" wording)	—	No logic; wording rule respected in code searched	NOT IMPLEMENTED	F-26
BP-5.1a	BAB 5.1	Inputs: coordinates, thickness, modulus, CBR, axle load, precipitation	`types/simulation.ts:6-19`	No spatial coordinates input; scalars only	PARTIALLY VERIFIED	F-09
BP-5.1b	BAB 5.1	Four-term loss; Burmister PDE residual at collocation points	—	—	NOT IMPLEMENTED	F-01
BP-5.1c	BAB 5.1	Adaptive weighting, Adam + L-BFGS	—	Only text "Konvergensi Adam Optimizer epoch 1500" (constant)	NOT IMPLEMENTED	F-02
BP-5.2	BAB 5.2	OpenVINO multi-device; YOLO on CPU/iGPU, PINN on iGPU/CPU, SLM on NPU	`deviceDetector.ts`, `simulatedBenchmark.ts`	Simulated; false NPU inference	NOT IMPLEMENTED	F-12, F-14
BP-6.0	BAB 6	Citizen (no login) + Field Engineer + Operations Manager + Policy Maker	`PUBLIC_REPORTER/VIEWER`, `FIELD_ENGINEER`, `OPERATIONS_MANAGER`, `POLICY_MAKER`	RBAC tests; server enforcement	VERIFIED (demo identity)	Supabase Auth BLOCKED
BP-6.1	BAB 6 role 1	Engineer one-click approve + SPK	Engineer recommends; Operations Manager approves; internal work order "bukan SPK"	Separation-of-duty tests	DIVERGENT	Deliberate (safer) — record against blueprint
BP-6.2	BAB 6 role 2	Alert widget, dynamic tonnage, Mobile WIM deployment	`pinnEngine.ts:135-170`; WIM views (SIMULASI)	Heuristic thresholds; simulated plates	PARTIALLY VERIFIED	F-06; enforcement wording
BP-6.3	BAB 6 role 3	5-year PCI chart, APBN savings, anomaly alert, official PDF	Policy views (ILUSTRATIF); `pdfGenerator.ts`	Illustrative figures; PDF without disclaimer	PARTIALLY VERIFIED	F-19, F-22
BP-7.1	BAB 7.1	Dark mode, glassmorphism, semantic colours	`src/styles`, components	Visual only; not audited in depth	NOT VERIFIED	—
BP-7.2a	BAB 7.2	Isometric 3D map (pitch 45-60°) of the corridor	`MapLibreMap.tsx`, `LeafletMap.tsx`	No pitch setting found	NOT IMPLEMENTED	F-15

Req. ID	Blueprint ref.	Expected behaviour	Implementation location	Evidence	Status	Gap
BP-7.2b	BAB 7.2	Pulsing pin on new report via Supabase Realtime	Map markers; no realtime channel	No EventSource/WebSocket/realtime code	PARTIALLY VERIFIED	F-24
BP-7.2c	BAB 7.2	Smart pop-up with YOLO evidence, live context, Intel telemetry badge	Report sheets, map popups	Constants / simulated values	PARTIALLY VERIFIED	F-02, F-05
BP-7.2d	BAB 7.2	Holographic 3D cross-section (Three.js), stress wave, pore-pressure particles	`SimulationPanel.tsx:460-517` (2D divs)	No Three.js dependency	NOT IMPLEMENTED	F-15, F-29
BP-7.2e	BAB 7.2	GenAI narrative box per role	`narrativeService.ts`; `explain.ts`	Cloud LLM or template; deterministic explanation on new surfaces	PARTIALLY VERIFIED	F-15
BP-7.2f	BAB 7.2	Route navigation buttons	"Lihat di peta" only (G-SW-07)	—	NOT IMPLEMENTED	—
BP-7.2g	BAB 7.2	Approve & send SPK in real time	Workflow transitions persisted	Golden path test	PARTIALLY VERIFIED	Real-time push absent; SPK replaced by internal order
BP-7.2h	BAB 7.2	Export PDF / JSON	`pdfGenerator.ts` (jsPDF)	Generates PDF	PARTIALLY VERIFIED	F-19
BP-7.2i	BAB 7.2	What-If sliders auto-filled; real-time update of cross-section, strains, RUL	`whatIf.ts`, `SimulationPanel.tsx`, road detail	Implemented over the rule engine; persisted	PARTIALLY VERIFIED	Outputs inherit F-03/F-07
BP-8.1	BAB 8.1	Next.js 14/15, Tailwind, Framer Motion, react-leaflet/mapbox, three/r3f, Supabase, FastAPI + PyTorch + Ultralytics + OpenVINO	Vite 6 + React 19 + Express + Tailwind 4 + motion + Leaflet/MapLibre; no three, no Python	`package.json`	DIVERGENT	F-15
BP-8.2a	BAB 8.2	`road_segments` with asphalt type, thickness, modulus, CBR, WIM load, rainfall, PCI	`road_segments` + `pavement_baselines` + observation tables (migration, not applied)	Schema normalised differently	DIVERGENT	F-24
BP-8.2b	BAB 8.2	`damage_reports` incl. `is_spoofed`, strains, RUL, pore pressure, sub-standard flag, role recommendations	`report_cases`, `damage_detections`, `pinn_predictions` (jsonb outputs)	No `is_spoofed`, no pore pressure, no sub-standard flag	DIVERGENT	F-17, F-24, F-26
BP-8.2c	BAB 8.2	`what_if_simulations` incl. `inference_latency_ms`	`simulations` table; file store records	Persisted (test); no latency field	PARTIALLY VERIFIED	Latency must not be stored until measured
BP-9	BAB 9	Integrated flow Citizen → AI pipeline → OpenVINO → GenAI → RBAC dashboards	Workflow + rule engine + dashboards	Workflow verified; AI stages absent	PARTIALLY VERIFIED	Figure 2.2
BP-10.1a	BAB 10.1	RMSE, MAE, MAPE, $R^2 > 0.90$ vs FWD/FEM; mAP for YOLO	—	No evaluation code or data	NOT IMPLEMENTED	F-16, F-27

Req. ID	Blueprint ref.	Expected behaviour	Implementation location	Evidence	Status	Gap
BP-10.1b	BAB 10.1	Latency < 10 ms, throughput, power on CPU/iGPU/NPU	—	—	NOT IMPLEMENTED	F-14
BP-10.1c	BAB 10.1	PCI prediction error, overlay accuracy, alert TP > 95 %, SUS > 80	—	—	NOT IMPLEMENTED	Requires field data and user study
BP-10.2	BAB 10.2	Roadmap 2026-2030	—	Narrative	N/A	—
BP-10.3a	BAB 10.3	Mitigation: virtual WIM + synthetic baseline + physics loss accurate with little data	Synthetic baseline only	"Accurate with little data" untestable without a PINN	NOT VERIFIED	F-01
BP-10.3b	BAB 10.3	Mitigation: anti-spoofing at the edge	—	—	NOT IMPLEMENTED	F-17
BP-10.3c	BAB 10.3	Mitigation: 100 % offline inference	Offline static-host mode for the UI	No offline model inference exists	NOT VERIFIED	F-14
BP-10.3d	BAB 10.3	Mitigation: slider ranges locked to physical values	`assessment.ts` ranges; `intelligence.routes.ts` 422	Implemented on new API; legacy `/api/simulate` unvalidated	PARTIALLY VERIFIED	F-11
BP-10.3e	BAB 10.3	Mitigation: neutral "anomaly" wording	Wording respected where searched	Sub-standard alert itself absent	PARTIALLY VERIFIED	F-26

Table D.2 — Traceability totals (67 rows, counted from Table D.1)

Status	Count
VERIFIED (software, demo identity)	3
PARTIALLY VERIFIED	24
DIVERGENT	7
NOT IMPLEMENTED	26
NOT VERIFIED	4
N/A	3

Appendix E — Data and Model Inventory

Table E.1 — Model and artifact inventory

Artifact	Expected	Found	Location	Status
PINN definition / training code	Python module	none	—	NOT IMPLEMENTED
PINN weights / checkpoint	`.pt`, `.onnx`, IR `.bin`	none	—	NOT IMPLEMENTED
CV weights	YOLOv8 `.pt` / IR	none	—	NOT IMPLEMENTED
CV training notebook	notebook	external Kaggle notebook (not in repository)	`FILES/Docs/` (parent folder)	Candidate; not reproducible here
OpenVINO IR	`.xml` + `.bin`	template XML string	`server/services/openvinoExportService.ts`	FAILED
Formula engine	—	`pinnEngine.ts`	`src/engines/pinn/core/`	deterministic, 296 lines
Golden regression fixture	—	76 scenarios	`tests/physics-validation/fixtures/pinn-golden.json`	regression only
Reference solver (audit)	—	MLET on-axis solver + verification + comparison	`docs/audit/forensic-2026-10-14/evidence/`	VERIFIED vs closed forms
Road datasets	surveyed attributes	hand-authored TypeScript constants	`src/data/reference/*.ts`	synthetic

Table E.2 — Dataset requirement for closure

Claim to support	Minimum data	Split rule
Strain response	layer thicknesses and moduli (incl. base, sub-base), ν, wheel load and pressure, plus FWD basins or instrumented sections	calibrate on some sections, validate on others never used for calibration
Moisture effect	soaked/unsoaked CBR or seasonal FWD on the same sections, drainage condition	seasonal hold-out
Deterioration / RUL	≥ 2 dated PCI surveys per segment, traffic and rainfall between them, maintenance history	temporal hold-out
Load	WIM axle spectra or axle-load surveys	station hold-out
CV	annotated local photos with consent and location metadata policy	location-disjoint split; near-duplicate removal

Appendix F — Benchmark Methodology

No benchmark was executed (no model, no runtime). Any future benchmark must report: hardware (CPU model, iGPU, NPU, driver versions, power mode), OS, OpenVINO version and `available_devices`; model hash and source; precision (FP32/FP16/INT8/INT4) and NNCF configuration with calibration set; input shape and batch size; 50 warm-up runs; at least 1000 timed runs with p50/p95/p99 and mean ± standard deviation; cold-start (first inference including compilation) separately; throughput at the reported batch; peak resident memory; device utilisation if measured with a named tool; numerical deviation (max absolute and relative) of each precision and device against FP32 on a fixed validation set and, for physics models, against the MLET reference; the exact script and command. Simulated numbers must never be entered in the register.

Appendix G — Remediation Acceptance Criteria (critical and high findings)

Table G.1 — Closure conditions

Finding	Closure condition (all must hold)
F-01	(a) grep for "Physics-Informed Neural Network", "PINN Solver", "neural", "PDE solver" in `src/`, README and the PDF template returns only roadmap or status text; or (b) model card, training script, data split, checkpoints and held-out metrics exist and the UI loads that model
F-02	No `Math.random` in engine, telemetry, benchmark, CV or attribution code; every displayed latency/loss/power figure links to a stored measurement record
F-03	Automated comparison with a verified MLET reference over a documented grid: relative error ≤ 15 % (or a justified tolerance) for εt, εv, w_0; similarity exponent −1 ± 0.01; reference cross-checked with a certified program (R-P3-01)
F-04	Modal removed or reduced to a labelled mock-up without numbers or official-source wording
F-05	No model name or confidence shown unless `modelStatus = REAL` with a recorded `modelVersion` and published metrics
F-06	Threshold register complete with document and clause per value; one constants module; test proves engine, UI and PDF agree
F-07	Deterioration model documented; life responds to structural inputs in documented directions; one failure definition; finite outputs for all valid inputs
F-08	One standard axle constant, cited; cross-module test
F-09	Input contract with units for load, pressure/radius, configuration, all layers' E, ν, h; dimensional note
F-10	Each moisture/temperature relation cited or calibrated with validity range and uncertainty
F-11	Contract tests show 400/422 for invalid, missing, non-numeric, negative and out-of-range inputs; no non-finite output
F-12	Hardware reported only from a positive device query; unit test without WebGL info returns "unknown"
F-13	Export endpoints return "not available" or are captioned templates until a converted model exists
F-14	Benchmark register filled from a reproducible script per Appendix F
F-15	Decision record; matrix shows no unexplained DIVERGENT rows
F-16	Data inventory with provenance, licence and status for every dataset; validation badges only with records
F-17	Anti-spoofing detector with FAR/FRR on a held-out set; rejected reports retained for audit
F-18	Recommendation from a cited design method; inputs shown; "advisory — not a construction design"
F-19	Disclaimer and status on every PDF page; no institution named without authorisation
F-20	No attribution percentage without a documented decomposition method

Appendix H — Glossary

Table H.1 — Glossary

Term	Definition as used in this report
PINN	A neural network u_θ trained with a loss that includes residuals of governing equations evaluated by differentiating u_θ, plus data and boundary terms
PDE	Partial differential equation; here the equations of elastostatics
MLET / Burmister	Multilayer linear-elastic theory for axisymmetric loads on bonded or unbonded horizontal layers over a half-space
Boussinesq solution	Closed-form elastic solution for a homogeneous half-space; the single-layer limit of MLET
Elastic similarity	Property that scaling all moduli by k scales all strains and displacements by 1/k
FWD	Falling Weight Deflectometer: impulse load test measuring surface deflection basins
FEM	Finite Element Method
CBR	California Bearing Ratio, a penetration index of soil strength (%)
ESAL	Equivalent Single Axle Loads (80 kN standard axle)
PCI	Pavement Condition Index, 0–100, computed from surveyed distress
RUL	Remaining Useful Life
Transfer function	Empirical relation between a critical strain and the number of load repetitions to failure
OpenVINO	Intel toolkit and runtime for optimised inference on Intel CPU, GPU and NPU
IR	OpenVINO Intermediate Representation (.xml topology + .bin weights)
NNCF	Neural Network Compression Framework (quantisation, pruning)
iGPU / NPU	Integrated graphics processor / neural processing unit
INT8 / FP16 / INT4	Numerical precisions used for quantised inference
WIM	Weigh-In-Motion axle-load measurement
ODOL	Over Dimension Over Loading
SLM	Small language model
RBAC	Role-based access control
Provenance	Record of where a value came from and its status (observed, inferred, synthetic, simulated)